Quantum Synthetic Data Generation for Industrial Bioprocess Monitoring

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Abstract

Data scarcity in biomanufacturing constrains model development for process monitoring and optimization. We evaluate a Quantum Wasserstein Generative Adversarial Network with Gradient Penalty (QWGAN-GP), whose generator is a Parameterized Quantum Circuit (PQC), against parameter-matched classical adversarial baselines on a single laboratory-scale photobioreactor cultivation. On this single cultivation, under a matched parameter budget, 2000 training epochs, and n=5 seeds, the quantum WGAN-GP cluster dominates the parameter-matched classical adversarial WGAN cluster on all four matched-budget fidelity metrics: log-return and optical-density temporal alignment (LR-DTW and OD-DTW), and log-return and optical-density single-step marginals (LR-EMD and OD-EMD). The optical-density DTW cluster separation is Welch-significant under family-wise Bonferroni correction; the optical-density EMD separation is Welch-significant uncorrected but does not survive Bonferroni (Supplementary §A.4). The quantum-cluster mean lag-1 autocorrelation is also closer to the real reference than any classical-baseline mean, with per-seed overlap noted. Per-model dominance is not extended to per-seed claims on every axis. Specific values appear in Section 4 and Table 2. We present matched-budget evidence that a PQC generator can match or exceed parameter-matched classical adversarial baselines on data-scarce bioprocess time-series synthesis, with potential relevance to soft-sensor and control applications.

Topical Heading: Process Systems Engineering

Keywords: quantum machine learning, Generative adversarial network, Bioprocess monitoring, Synthetic time series generation, Data augmentation

Plain Language Summary

On a small bioprocess dataset, a quantum generator reproduced both the per-step value distribution and the short-term temporal pattern more faithfully than equally-sized classical WGAN baselines on every fidelity metric we measured.

1 Introduction

Machine Learning (ML) and Artificial Intelligence (AI) have emerged as transformative tools in diverse scientific disciplines, from cancer research to financial analysis (Figure 1) [1, 2, 3, 4]. However, complex systems such as manufacturing, drug discovery, and formulated products [5] have lagged in fully embracing ML/AI methodologies compared to other fields. This discrepancy reflects unique challenges including high dimensionality, inherent variability, and complex underlying chemical, physical, or biological mechanisms [6].

Bioprocess engineering exemplifies these challenges. The field is crucial for producing biofuels, pharmaceuticals, and sustainable materials, yet data analytics applications often grapple with limited datasets due to time-consuming and expensive experiments [7]. This scarcity hinders development of accurate predictive models, particularly for hard-to-quantify quality indicator variables (QIVs) essential for process optimization and real-time monitoring.

Critical QIVs in bioprocesses include biomass concentration, which is often esti-

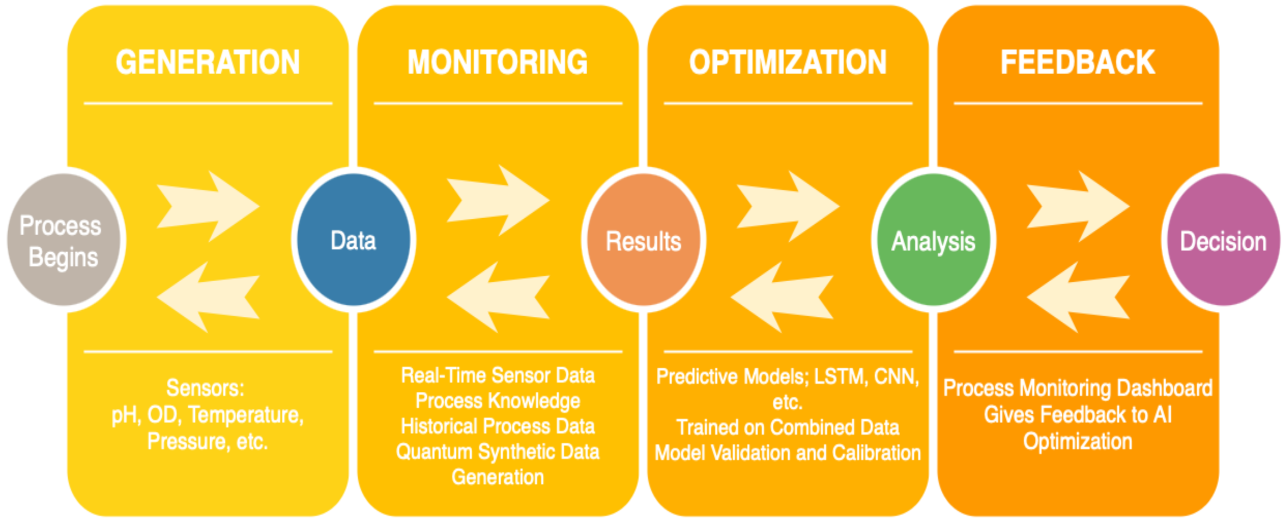

Figure 1: Conceptual diagram of how we envision using quantum generative AI within industrial bioprocess engineering.

mated through optical density measurements or viable cell counts but requires complex calibration and suffers from interference in high-density cultures [8]. Product concentration and quality, such as monoclonal antibody titers and glycosylation patterns in mammalian cell culture, typically require offline analytical methods such as High-Performance Liquid Chromatography (HPLC) or mass spectrometry. Metabolic activity indicators, including specific growth rates, substrate consumption rates, and metabolic flux distributions, are fundamental QIVs that are difficult to quantify in real-time without sophisticated analytical platforms. The integration of ML with bioprocess engineering has been gradually expanding, with applications spanning process monitoring [9], optimization [10], and control [11]. These approaches have demonstrated potential to address limitations of traditional mechanistic models, particularly when dealing with non-linear dynamics and multivariate nature of biological systems.

1.1 Data Scarcity as a Fundamental Limitation

A fundamental challenge limiting widespread adoption of machine learning in bioprocess engineering is data scarcity [12, 13]. Unlike fields such as computer vision or natural language processing, where datasets may be comprised of millions of samples, bioprocess datasets are typically constrained by several factors. Bioprocess experiments, particularly on pilot and production scales, involve substantial costs in terms of equipment, raw materials, personnel, and time [14, 15]. A single bioreactor run at pilot or production scale may take weeks to complete and can be costly, often on the order of several hundreds of thousands of dollars or more, severely limiting the number of experimental conditions that can be feasibly explored [15]. Biological systems exhibit inherent stochasticity and sensitivity to environmental conditions, resulting in significant batch-to-batch variability even under nominally identical operating conditions. Commercial bioprocesses represent significant intellectual property, which results in limited data sharing across the industry. While fields such as computer vision benefit from open datasets with millions of examples, bioprocess data typically remains siloed within individual companies or research groups. These factors collectively result in datasets that are often too small to effectively train complex ML models, particularly deep learning architectures that typically require substantial amounts of diverse training data.

1.2 Synthetic Data Generation Approaches

Synthetic data generation is a promising approach to address data scarcity challenges in bioprocess engineering [16]. By creating artificial datasets that faithfully represent statistical properties and dynamic behaviors of real bioprocesses, synthetic data approaches can potentially overcome limitations in data availability while avoiding confidentiality concerns (a comparison of approaches is provided in Supplementary Table A1

First-principles models have been used to simulate bioprocess dynamics under various conditions, generating synthetic time series data [17]. Although this approach ensures physical consistency, its fidelity is limited by the accuracy of the underlying model and its ability to capture process variability and uncertainty. Methods such as Gaussian process regression [18] and multivariate statistical models can generate synthetic data based on estimated probability distributions. These approaches excel at reproducing statistical properties, but often struggle to capture complex temporal dependencies and rare events.

Machine learning models such as variational autoencoders (VAEs) and generative adversarial networks (GANs), including time-series-specific architectures [19], can learn directly from available data to generate synthetic samples [20]. These approaches offer flexibility in generating diverse datasets without requiring detailed mechanistic understanding, although their performance is highly dependent on the quality and quantity of training data. Combining mechanistic knowledge with data-driven techniques represents a promising direction for generation of synthetic bioprocess data.

Despite these advances, significant challenges remain in generation of high-quality data: accurate capture of complex temporal dependencies, incorporation of mechanistic knowledge constraints, ensuring physical realizability, and validating the quality of synthetic data. Addressing these challenges requires novel methodological approaches. These limitations motivate exploring alternative computational paradigms. Quantum approaches *may* offer advantages for certain structured learning tasks in chemical and biochemical engineering [21].

1.3 Quantum Generative Adversarial Networks

Quantum Generative Adversarial Networks (QGANs) embed Parameterized Quantum Circuits within the generator of a GAN [22, 23, 24]. This motivates the falsifiable question that frames the present study: *can a PQC generator match or exceed a parameter-matched classical generator on a low-data bioprocess task?* The PQC operates in an exponentially large Hilbert space with $\mathcal{O}(\text{poly}(n))$ parameters; whether that translates into a practical advantage is what we test. Prior work has reported QGAN results in low-data regimes and on multimodal distributions [25, 26], with applications in finance [27] and entangling-state generation [28]; GANs more broadly, including classical recurrent variants, have also been applied to healthcare time series [29]. We test this claim directly under a matched-parameter, matched-epoch protocol. Current quantum hardware constraints, noise, decoherence, and limited circuit depth, restrict practical scalability

[30], but QGANs offer a candidate alternative when classical GANs struggle with intricate temporal dependencies in limited datasets [31, 32]. A detailed review of QGANs and their relationship to classical GANs is provided in Section 2.

1.4 Principal Contributions

The principal contributions of this work are:

- **Matched-Parameter Comparison Protocol:** A parameter-matched, matched-epoch, multi-seed protocol for comparing a PQC generator against size-matched classical baselines on a data-scarce bioprocess task.
- **QWGAN-GP for Process Data Synthesis:** A hybrid QWGAN-GP architecture, with a Parameterized Quantum Circuit generator and a classical critic, evaluated on a laboratory-scale photobioreactor cultivation.
- **Matched-Budget Empirical Finding:** Under the matched protocol, the quantum cluster per-model means dominate the parameter-matched classical WGAN cluster on all four matched-budget fidelity metrics (log-return and optical-density temporal alignment, LR-DTW and OD-DTW; log-return and optical-density single-step marginals, LR-EMD and OD-EMD), with Welch significance on both optical-density axes (OD-DTW after family-wise Bonferroni correction; see Section 4.1 / supp §A.4); specific values are reported in Section 4.1 and Table 2, with explicit n=5 power caveats and per-seed-vs-per-model scoping preserved throughout.
- **Reproducible Open-Science Artifact:** We release the code, dataset, and matched-budget reproduction script under an open licence to enable independent verification and future studies of quantum generators for bioprocess time series.

A decision-tree triage workflow is also outlined in the Outlook as an organising concept for future work; it is not implemented or evaluated here and is not an empirical contribution of this study.

The remainder of this paper is structured as follows. Section 2 reviews classical and quantum machine learning foundations. Section 3 describes the QWGAN-GP architecture and the photobioreactor experimental setup. Section 4 presents results, key contributions, and discusses implications and limitations. Section 5 offers concluding remarks.

2 Classical and Quantum Machine Learning

This section reviews the classical and quantum machine learning foundations underpinning our approach, beginning with GANs and the WGAN-GP formulation before turning to quantum machine learning and QGANs.

2.1 Generative Adversarial Networks and WGAN-GP

GANs represent the foundation of modern synthetic data generation [31]. The GAN framework (illustrated in Supplementary Figure A1 consists of two neural networks engaged in a competitive minimax game: a generator G that learns to produce synthetic data, and a discriminator D that distinguishes between real and synthetic samples. The objective function is formulated as:

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_{data}(x)}[\log D(x)] + \mathbb{E}_{z \sim p_z(z)}[\log(1 - D(G(z)))] \quad (1)$$

where $p_{data}(x)$ represents the real data distribution and $p_z(z)$ is the noise distribution.

For time-series applications, specialized architectures such as TimeGAN [19] incorporate recurrent components to better capture temporal dependencies. However, classical GANs remain constrained by training stability issues, limited ability to capture multimodal distributions, and sensitivity to hyperparameter selection [33, 34], limitations that become pronounced with the small datasets typical of bioprocess engineering.

Despite their success, the original GAN formulation suffers from mode collapse and vanishing gradients [33]. The Wasserstein GAN (WGAN) addressed these instabilities by replacing the Jensen-Shannon divergence with the Earth Mover's distance [35], but its reliance on weight clipping introduced pathological behavior. The Wasserstein GAN with Gradient Penalty (WGAN-GP) [34] refined this by imposing a soft gradient constraint:

$$\min_G \max_D \big[\mathbb{E}_{x\sim P_r}[D(x)] - \mathbb{E}_{z\sim P_z}[D(G(z))] - \lambda \mathbb{E}_{\hat{x}\sim P_{\hat{x}}}\big[(\|\nabla_{\hat{x}} D(\hat{x})\|_2 - 1)^2\big]\big] \tag{2}$$

The WGAN-GP framework employs a critic network D(·) and generator network G(·) in a minimax optimization scheme. The objective approximates the Wasserstein distance between the real data distribution P_r and the distribution of generated samples produced from noise distribution P_z. To enforce the Lipschitz constraint, a gradient penalty term is introduced with coefficient λ(typically set to 10), which operates on interpolated samples $\hat{x} = \epsilon x + (1-\epsilon)G(z)$ where $\epsilon \sim \text{Uniform}(0,1)$. These interpolated points, distributed according to $P_{\hat{x}}$, provide a differentiable path between real and generated samples, enabling stable enforcement of the 1-Lipschitz continuity requirement without weight clipping.

This solution stabilizes training across architectures and mitigates issues with weight clipping, leading to improved sample quality and more reliable convergence [36]. A detailed mathematical derivation of the Wasserstein distance and the Kantorovich-Rubinstein duality is provided in Supplementary Section A.2.2.

2.2 Quantum Machine Learning and QGANs

Quantum machine learning (QML) leverages quantum mechanical principles to improve learning algorithms, with recent advances demonstrating potential advantages in expressivity, training efficiency, and generalization [37, 38]. Relevant approaches include quantum-enhanced sampling, quantum kernels [39, 40], and variational quantum algorithms, each offering potential benefits for data-scarce bioprocess applications [30].

QGANs are a hybrid quantum-classical approach in which a Parameterized Quantum Circuit (PQC) serves as the generator, first introduced by Dallaire-Demers and Killoran [22]. Superposition and entanglement give a PQC access to a state space that grows exponentially with qubit count; whether this translates into a practical advantage over a parameter-matched classical generator is an empirical question that remains open and is one we examine directly in this work rather than assume [25]. QGANs have been applied to finance [27] and entangling-state generation [28]; classical recurrent GAN variants have also been applied to healthcare time series [29]. For time-series generation, models such as TimeGAN [19] and DoppelGANger [41] have shown strong classical performance, but remain limited by computational constraints with high-dimensional distributions, a gap that quantum approaches may help address.

3 Methods

3.1 QWGAN-GP Architecture Overview

Our QWGAN-GP employs a hybrid quantum-classical architecture in which the generator is a Parameterized Quantum Circuit (PQC) and the critic is a classical convolutional neural network trained under the WGAN-GP objective (Eq. 2). The quantum generator uses 5 qubits arranged in three layers, incorporating Hadamard gates for superposition, IQP encoding with parameterized R_z rotations, two strongly entangling layers with $\mathrm{Rot}(\phi, \theta, \omega)$ gates, and range-based CNOT gates for entanglement. The circuit outputs 10-dimensional samples via Pauli-X and Pauli-Z expectation values, matching the rolling window subsequence length. The generator contains 55 trainable parameters. Full circuit specifications and the critic architecture are detailed in Supplementary Sections A.5 and A.8, respectively (see also Figure A5).

The raw optical density time series was transformed to log returns, standardized to zero mean and unit variance, and linearly rescaled to $[-1, 1]$ using the global min and max of the standardized log-return series, prior to training (Pipeline B; details and the ablation that motivated its selection are in Supplementary Section A.7). Overlapping subsequences of length 10 with stride 2 were extracted using a rolling window approach [42].

The WGAN-GP loss formulation (Eq. 2) was chosen for the quantum generator because it provides stable gradients and meaningful convergence metrics, and mitigates mode collapse issues that can be exacerbated by the discrete nature of quantum measurements; we do not attribute reduced mode collapse to the quantum generator.

Training protocol (matched-budget contract). All adversarial models in this study are trained under a single matched-budget contract so that, within the WGAN-GP adversarial cohort, the only component that differs between the quantum and classical entrants is the generator (the VAE and AR(2) non-adversarial baselines have their own native training procedures, detailed below). The contract is: 2000 training epochs (no early stopping during matched-budget reproduction), 5 independent random seeds $\{42, 43, 44, 45, 46\}$, a shared 250881-parameter classical critic used identically across all WGAN-GP runs, Adam with $\beta_1 = 0.0$ and $\beta_2 = 0.9$, generator learning rate $\eta_G = 6.9173 \times 10^{-5}$, critic learning rate $\eta_C = 1.8046 \times 10^{-5}$, $n_{\text{critic}} = 9$ critic updates per generator update, gradient-penalty coefficient $\lambda_{\text{GP}} = 2.16$, and batch size 12. The four quantum ansatz variants (the 55-parameter `iqp_sel_55` reproduction and three `default_75`-family circuits V1, V2, V3 at 75, 135, and 75 parameters respectively), the three classical adversarial baselines (`wgan_mlp`, `wgan_cnn`, `wgan_lstm`), and the two non-adversarial baselines (AR(2), VAE) are all trained under the same 2000-epoch, 5-seed budget. AR(2) is fit in closed form via least squares; the VAE uses a single-Adam ELBO loop ($\eta = 1 \times 10^{-3}$) and does not consume the shared critic or gradient-penalty terms. The remaining hyperparameter values are hyperparameter-optimization-locked from the v1.0 release (frozen prior to this revision); the matched-budget reproduction inherits them unchanged so that all 2000-epoch comparison runs share identical optimizer machinery. Full per-model dtype and backend pinning details are given in Supplementary Section A.8 and Appendix.

Note on hyperparameter selection. The shared adversarial hyperparameters listed above (η_G, η_C, n_{critic}, λ_{GP}, batch size) were selected during quantum-generator hyperparameter optimization for the IQP:SEL baseline and applied unchanged to the parameter-matched classical WGAN-GP comparators; the classical baselines were not separately re-tuned. This is a deliberate matched-protocol choice: re-tuning per architecture would conflate generator-architecture effects with HPO effects under the cross-model comparison. As a consequence, the reported classical-cluster metric values reflect classical performance under quantum-selected hyperparameters rather than under classical-optimal HPO, and the WGAN cluster numbers should be read accordingly.

3.2 Circuit Design Rationale

The generator circuit was fixed by three deliberate design choices, each constrained by the matched-parameter, low-data setting of this study.

Qubit count. The generator uses 5 qubits. The qubit count is dictated by the rolling-window length: each length-10 subsequence is produced from Pauli-X and Pauli-Z expectation values over the 5-qubit register (5 qubits $\times\, 2$ observables $=$ 10 outputs), so the window length and qubit count are coupled by construction rather than tuned independently. Five qubits give a 2^5-dimensional state space, which is more than sufficient to embed a length-10 log-return window while keeping the parameterised circuit trainable by exact backpropagation on a state-vector simulator at the data scale available here. The locked configuration therefore has 5 qubits, 3 layers, and 55 trainable parameters, decomposed as one IQP-encoding parameter per qubit plus 3 strongly-entangling rotation parameters per qubit per layer.

Ansatz expressibility versus trainability. The reported circuit (IQP encoding + strongly-entangling layers, range entangler, 55 parameters) was selected against a multi-seed ansatz comparison at a matched 2000-epoch budget. We compared three alternatives at increased capacity or different connectivity: a depth-4 range-entangled circuit (75 parameters), a depth-8 range-entangled circuit (135 parameters), and a depth-4 linear-entangled circuit (75 parameters), each over 5 seeds. Increasing parameter count and circuit depth, or changing the entangler topology, did *not* yield a fidelity improvement over the compact 55-parameter circuit at the matched budget on this dataset (per-model values reported in Section 4; full per-model numbers are released in the artifact set, see Data Availability). Consistent with the expressibility-

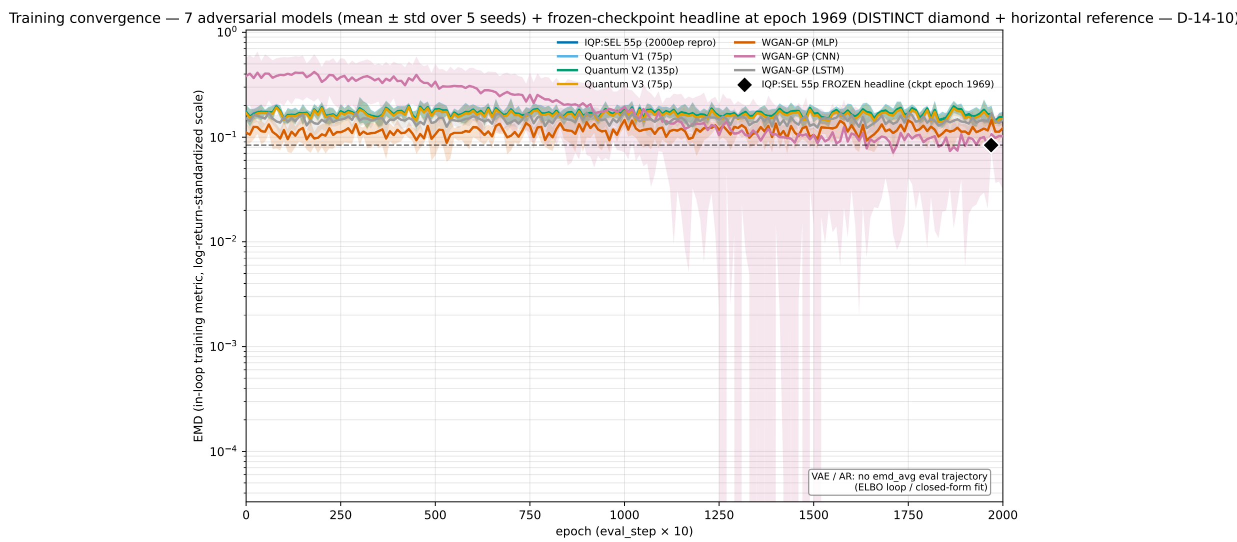

Figure 2: Matched-budget training convergence: OD-scale in-loop EMD versus training epoch (log-y) for the 7 WGAN-GP generators (4 quantum + 3 classical adversarial). Curves are 5-seed mean $\pm$ sample standard deviation over seeds $\{42, 43, 44, 45, 46\}$, with EMD evaluated every 10 training epochs across the 2000-epoch budget. The frozen-headline reference (epoch-1969 checkpoint) is shown as a distinct diamond marker plus horizontal line. All 7 models converge to a tight cluster of mean OD-EMD ≈ 0.026 by $\sim$epoch 250 and remain in that band for the remainder of training, demonstrating that the matched-budget contract is held without divergence or collapse. The visible deep dip is the wgan_cnn seed-42 outlier (single seed where OD-EMD transiently fell to $\sim 10^{-4}$ before recovering, disclosed at face value rather than excluded). VAE and AR(2) are not plotted: VAE trains with a single-Adam ELBO loop (no per-epoch EMD eval was captured) and AR(2) is fit in closed form via `lstsq` (no epoch trajectory exists). Per-figure data provenance is summarised in the Data Availability section.

trainability tradeoff for variational circuits [43, 44], the deeper/larger ansatz is harder to optimise in the low-data, fixed-epoch regime without a corresponding accuracy gain, motivating the choice of the smaller circuit as the reported configuration. The 55-parameter design also sits well inside the regime where gradient-magnitude attenuation ("barren plateaus") is not a dominant concern, which is relevant context for the matched-budget dominance reported in Section 4 against parameter-comparable classical baselines.

Classical critic with a quantum generator. The critic is a classical convolutional network and only the generator is quantum. This is deliberate: the WGAN-GP objective requires a gradient penalty on interpolated samples, which is cheap and stable to evaluate with a classical critic, whereas placing the discriminator on the quantum device would add measurement-shot noise to the critic gradient and substantially increase circuit evaluations per step without benefiting the quantity of interest (the generated distribution). A classical critic also makes the comparison against the matched classical WGAN-GP baselines clean: within the adversarial cohort, the only component that differs between the quantum and classical entrants is the generator, isolating the effect of the quantum circuit.

3.3 Photobioreactor Experimental Setup

Time-series data were collected from a 20-liter photobioreactor (LUCY®, Synoxis Algae) designed for laboratory- and pilot-scale cultivation of microalgae; the system is also available in a larger 300-liter configuration. The 20-liter unit used in this study is shown schematically in Supplementary Figure A7 (Figure A7). The system comprises six vertical borosilicate glass tubes (6 cm OD, 120 cm length) connected in parallel, with SALT technology (spiral air lift tubular circulation) for gentle mixing, controlled aeration, and programmable lighting managed by an integrated automated controller.

The system continuously monitors pH, temperature, optical density, dissolved oxygen, and pressure via feedback control loops, with data logged at 10-minute intervals by an internal data acquisition system. The OD sensor operates at 880 nm and reports values in arbitrary units based on logarithmic attenuation, calibrated against fresh culture medium.

Dataset and preprocessing. The study uses a single LUCY photobioreactor cultivation campaign comprising 778 raw optical-density time points logged at 10-minute intervals. First differencing into log-returns yields 777 log-return observations ($r_t = \ln \mathrm{OD}_t - \ln \mathrm{OD}_{t-1}$); log-returns are the natural rate variable for this signal, because for an exponentially growing culture $\mathrm{OD}_t \approx \mathrm{OD}_0\, e^{\mu t}$, so r_t is (up to the 10-minute sampling interval) the local specific growth rate μ_t, the quantity of physiological interest in cultivation monitoring. These are then standardized to zero mean and unit variance, then linearly rescaled to $[-1, 1]$ using the global min and max of the standardized log-return series. Overlapping subsequences of length 10 with stride 2 are then extracted with a rolling window, producing 384 training windows. We refer to this preprocessing chain (log-return $\rightarrow$ standardization $\rightarrow$ rescale to $[-1, 1]$) as **Pipeline B** (native preprocessing); it is the only preprocessing under which the matched-budget evaluation results reported in Section 4 are computed. Two alternative pipelines were evaluated and discarded prior to the matched-budget runs: *Pipeline A* (raw min-max-normalized OD; pre-v1.0 legacy normalization, retained solely for the historical reconciliation note) and *Pipeline C* (log-return $\rightarrow$ standardize $\rightarrow$ inverse Lambert W $\rightarrow$ rescale to $[-1, 1]$; i.e., Pipeline B with an inverse Lambert W heavy-tail correction inserted between standardization and rescaling, matching the v1.1 published pipeline). Pipeline C was dropped per decision D-10-05: a 5-seed ablation at 1000 epochs showed Pipelines B and C tied on OD-scale EMD (0.0276 vs 0.0261), OD-scale ACF lag-1 (0.696 vs 0.697), OD-scale DTW (0.30 vs 0.41), and TSTR-lite R^2 (0.994 vs 0.994), so Pipeline B was selected for simplicity and because it directly addresses reviewer concern R1-M3 that the Lambert W step over-Gaussianizes the input distribution. Supplementary Section A.7 reports the full ablation (Figure A9). Because only one independent campaign is available, all 384 windows are used for training; no held-out validation or test split is carved out, as a 384-window single campaign is too small to support a held-out split without severely under-powering training (stated openly per the calibration-honesty standard, R1-M5). Multi-campaign generalization is identified as future work. All reported quantitative results are aggregated over 5 independent random seeds (42, 43, 44, 45, 46) and reported as mean ± standard deviation.

4 Results and Discussion

4.1 Results

We evaluate the fidelity of the generated synthetic time series using four complementary metrics. Dynamic Time Warping (DTW) distance measures temporal alignment, capturing non-linear dependencies where process variations cause phase shifts in growth events. Probability density and cumulative distribution functions assess distributional fidelity. Quantile-quantile plots evaluate tail behavior. Autocorrelation functions verify that no spurious temporal structure is introduced. Together, these metrics cover the statistical properties most critical for downstream bioprocess modeling applications.

Evaluation scale. Every fidelity metric is reported on *both* the transformed log-return scale (the space the generator is trained in) and the original optical-density (OD) scale (obtained by inverting the preprocessing pipeline), so that the physical-unit fidelity is explicit (R1-m3). Table 1 states the scale of each metric family and the headline 55-parameter IQP:SEL quantum generator value on each scale (5-seed mean ± sample standard deviation under the matched 2000-epoch budget).

PDF/CDF, Q-Q, and ACF diagnostic plots in the main text are shown on the transformed log-return scale (the training space); the corresponding original-OD-scale plots are provided in the supplement. DTW, EMD, and the distributional moments are reported on both scales as above.

Cross-Model Comparison

The headline empirical finding of this study, summarised across all nine generators in Table 2, is that the quantum WGAN-GP cluster dominates the parameter-matched

Table 1: Evaluation metrics and the scale on which each is computed. Values are mean ± sample standard deviation over 5 seeds {42, 43, 44, 45, 46}, for the 55-parameter IQP:SEL quantum generator (`iqp_sel_55_repro`), matched 2000-epoch budget, Pipeline B preprocessing. Per-figure data provenance is summarised in the Data Availability section.

Metric	**Transformed (log-return)**	**Original (OD)**	**Reported on**
EMD	0.0040 ± 0.0009	0.0282 ± 0.0043	both scales
DTW (mean)	6.3339 ± 1.0936	0.3702 ± 0.0235	both scales
Moment: mean	0.1231 ± 0.0047	1.4039 ± 0.0082	both scales
Moment: std	0.8297 ± 0.1305	0.8783 ± 0.0117	both scales
Moment: skewness	-0.0015 ± 0.0283	1.3705 ± 0.0173	both scales
Moment: kurtosis	0.2039 ± 0.0555	0.8431 ± 0.0504	both scales
ACF (lag 1, mean)	-0.0949 ± 0.0092	0.4667 ± 0.0286	both scales

Table 2: Cross-model comparison on the five headline matched-budget metrics (5-seed mean; n=5 seeds {42, 43, 44, 45, 46}; Pipeline B; 2000 epochs). Models are grouped by family. Bold marks the row leader: lowest mean for EMD/DTW (lower-is-better), or the mean closest to the real-data lag-1 ACF reference (-0.064) for ACF lag-1. The VAE LR-DTW value (marked †) is excluded from the LR-DTW row leader because the VAE operates in a degenerate generation regime: the synthetic log-return series exhibits a strongly anti-correlated step-to-step structure (lag-1 ACF ≈ -0.65 vs real ≈ -0.064), producing a high-frequency oscillation that is warped-aligned to the real series at low DTW cost without carrying real temporal information (see “VAE characterization” paragraph below). For OD-EMD the quantum cluster mean (Q range 0.0279-0.0308) is significantly below the WGAN cluster mean (WGAN range 0.0769-0.7989) at Welch $p = 0.019$; bolding the lowest sample mean is an arithmetic convention applied within the non-adversarial column block and does not displace the quantum/WGAN cluster-level conclusion. Per-seed sample standard deviations and additional decimals are in Figures 3 and 4. Per-figure data provenance is summarised in the Data Availability section.

	Quantum				**Classical adversarial**			**Non-adversarial**	
Metric	IQP:SEL (55p)	V1 (75p)	V2 (135p)	V3 (75p)	WGAN-MLP (74p)	WGAN-CNN (73p)	WGAN-LSTM (78p)	VAE (562p)	AR(2) (3p)
LR-EMD	0.0040	0.0041	0.0044	0.0050	0.0444	0.1286	0.0244	0.0158	**0.0029**
OD-EMD	0.0282	0.0283	0.0279	0.0308	0.0769	0.7989	0.1177	**0.0257**	0.0291
LR-DTW	6.33	6.25	**6.09**	9.48	28.51	69.02	18.23	0.0876†	7.70
OD-DTW	0.3702	0.3491	**0.3329**	0.4096	0.9148	6.9907	0.5967	0.3069	0.3707
lag-1 ACF	−0.0949	−0.0997	−0.0968	**−0.0895**	−0.1418	−0.1112	−0.2422	−0.6482	−0.1356

†VAE LR-DTW excluded from row leader: the small value reflects a degenerate generation regime (lag-1 ACF ≈ -0.65 vs real ≈ -0.064 produces a high-frequency oscillation that is warped-aligned at low DTW cost), not faithful temporal alignment; see the “VAE characterization” paragraph below.

classical adversarial WGAN cluster on all four matched-budget metrics. On the log-return temporal-alignment axis (Dynamic Time Warping on the log-return scale, and the lag-1 autocorrelation of the log-return series), the four quantum WGAN-GP variants outperformed at the per-model-mean level the three parameter-matched classical adversarial baselines (mean-level dominance on LR-DTW and on lag-1 ACF closeness; per-seed overlap noted where present). On the log-return single-step marginal axis (LR-EMD), the quantum cluster also dominates the WGAN cluster on per-model means ($\approx$ 15$\times$ lower; Welch $p \approx 0.0002$); the AR(2) reference remains the lowest LR-EMD value at 0.0029, and the LR-EMD per-model dominance is not extended to a per-seed claim. On the optical-density single-step marginal axis (OD-EMD), the quantum cluster is below the WGAN cluster at Welch $p = 0.019$ (max $|d| = 2.34$). On the optical-density temporal-alignment axis (OD-DTW), the same direction holds and reaches Welch p as low as 0.002 against the strongest WGAN comparator. We report the four-metric quantum dominance against the WGAN adversarial cluster as the principal result, with explicit n=5 power caveats throughout, and with the AR(2) reference and VAE characterization preserved as parallel disclosures (AR(2) leads on LR-EMD; VAE LR-DTW reflects a degenerate generation regime).

On the log-return scale, the four quantum generators clustered at mean LR-DTW between 6.09 and 9.48 (V2= 6.09, V1= 6.25, iqp_sel_55_repro= 6.33, V3= 9.48), while the three classical adversarial baselines spanned 18.23 to 69.02 (wgan_lstm= 18.23, wgan_mlp= 28.51, wgan_cnn= 69.02) and the AR(2) reference fell at 7.70 (Figure 3). The worst-case quantum variant (V3, LR-DTW= 9.48) was below the best-case classical adversarial baseline (wgan_lstm, LR-DTW= 18.23), and no quantum-classical cell in the matched-budget grid showed a classical adversarial generator achieving a lower LR-DTW than any quantum variant. The VAE, with mean LR-DTW= 0.0876, is excluded from this comparison: its anomalously small DTW reflects a degenerate generation regime in which the synthetic log-return series exhibits a strongly anti-correlated step-to-step structure (lag-1 ACF ≈ -0.65 vs real

≈ -0.064; see VAE characterization below), producing a high-frequency oscillation that is warped-aligned to the real series at low DTW cost without carrying real temporal information.

On the log-return single-step marginal-distribution axis (LR-EMD, earth-mover's distance between the synthetic and real log-return empirical distributions), the cross-model ordering aligns with the LR-DTW ranking inside the WGAN-vs-quantum comparison, but the AR(2) reference now leads. Per-model LR-EMD means under the same matched-budget contract (Pipeline B, n=5 seeds) were: AR(2)= 0.0029, iqp_sel_55_repro= 0.0040, V1= 0.0041, V2= 0.0044, V3= 0.0050, VAE= 0.0158, wgan_lstm= 0.0244, wgan_mlp= 0.0444, wgan_cnn= 0.1286. On per-model means, every quantum variant (0.0040-0.0050) is lower than every classical WGAN adversarial baseline (0.0244-0.1286) on LR-EMD, with the quantum cluster mean roughly 15× lower than the WGAN cluster mean (Welch $p \approx 0.0002$ for the floor pair against WGAN adversarial baselines). The AR(2) reference leads the field at 0.0029, a factor of 1.4× lower than the lowest quantum variant. We report the quantum-vs-WGAN LR-EMD dominance as a between-cluster mean-level result with the n=5 power caveat preserved, and explicitly do not extend it to a per-seed dominance claim. LR-DTW and LR-EMD measure different fidelity axes (temporal-alignment cost under non-linear time warping versus single-step marginal-distributional distance); on this dataset the two rankings are now concordant inside the quantum-vs-WGAN comparison.

The lag-1 autocorrelation of the generated log-return series tells the same structural-alignment story as LR-DTW and is best read jointly with LR-DTW rather than as an independent metric. The real-data log-return lag-1 ACF, computed under the same matched preprocessing pipeline as the generators, was -0.064 (Figure 4, dashed reference). Every quantum variant produced a mean lag-1 ACF within a narrow band closer to the real reference than any other generator's mean: V3= -0.0895, iqp_sel_55_repro= -0.0949, V2= -0.0968, V1= -0.0997 (range -0.09 to -0.10). The classical adversarial baseline means overshot the negative lag-1

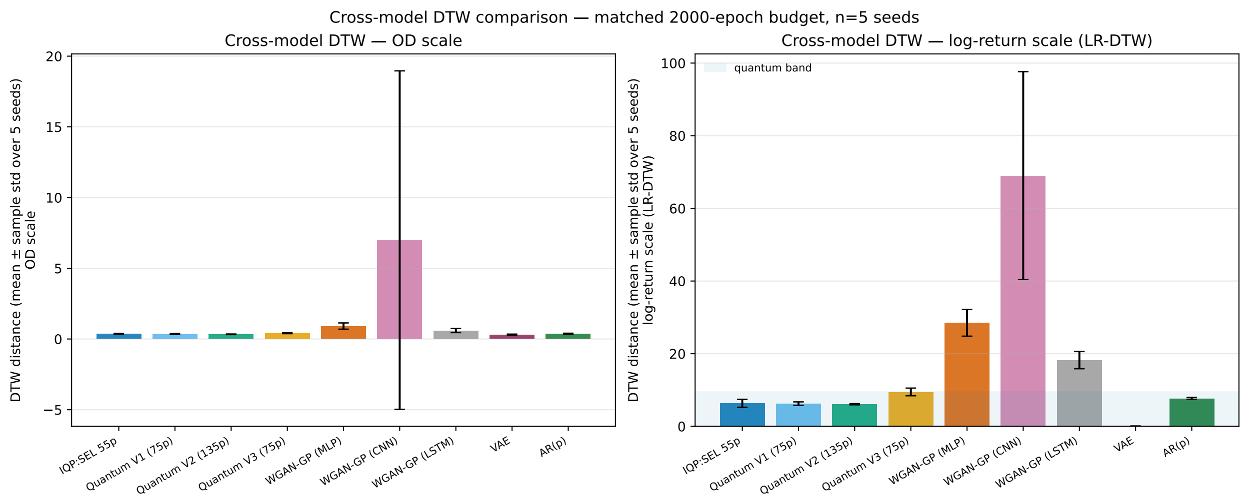

Figure 3: DTW distance under matched 2000-epoch training (n=5 seeds; mean ± sample standard deviation). Left: optical-density (OD) scale, on which the quantum cluster (V2= 0.33, V1= 0.35, iqp_sel_55= 0.37, V3= 0.41) is below the WGAN cluster (wgan_lstm= 0.60, wgan_mlp= 0.91, wgan_cnn= 6.99). Right: log-return (LR) scale, on which the quantum cluster (V2= 6.09, V1= 6.25, iqp_sel_55= 6.33, V3= 9.48) uniformly outperforms the classical adversarial baselines (wgan_lstm= 18.23, wgan_mlp= 28.51, wgan_cnn= 69.02) and falls below AR(2)= 7.70 for every quantum variant except V3 (V3= 9.48 > 7.70). VAE LR-DTW= 0.0876 reflects a degenerate generation regime (see text) and is excluded from the dominance comparison. Per-figure data provenance is summarised in the Data Availability section.

magnitude: wgan_cnn$= -0.111$, wgan_mlp$= -0.142$, wgan_lstm$= -0.242$; the AR(2) reference produced lag-1$= -0.136$. The VAE lag-1 ACF was -0.648, an order of magnitude larger in absolute value than the real-data reference and outside the cross-model spread on this metric. The mean-versus-mean comparison admits per-seed overlap: for example, the wgan_lstm seed-46 lag-1 ACF was -0.076, closer to the real-data reference -0.064 than any quantum-cluster mean. We therefore phrase the finding as a mean-level result: the quantum-cluster mean lag-1 ACF is closer to real than any classical-baseline mean, with per-seed overlap noted. The LR-DTW ranking and the lag-1 ACF mean-closeness-to-real ranking are concordant at the per-model-mean level: the quantum cluster leads on both, the classical adversarial cluster trails on both, and the VAE is anomalous on both. We treat the two observations as a single structural-alignment finding rather than as two independent metric outcomes.

VAE characterization. The VAE baseline does not fit cleanly into the cluster-level quantum-vs-WGAN comparison and we characterize it separately. On the OD-marginal it is well-aligned (OD-EMD $\approx$ 0.026); its LR-EMD$\approx$ 0.016 is also in-cluster with the other generators on the marginal axis, indicating that the per-step distribution is captured. However, its mean LR-DTW of 0.0876 is anomalously small because the synthetic log-return series exhibits a strongly anti-correlated step-to-step structure (lag-1 ACF ≈ -0.65, vs real ≈ -0.064). The resulting high-frequency oscillation is warped-aligned to the real series at low DTW cost despite carrying no real temporal information, so the small DTW does not reflect faithful temporal alignment. The lag-1 ACF -0.648, sharply different from the real-data -0.064, is the diagnostic that exposes the degenerate regime: the VAE produces a strongly anti-correlated step-to-step structure that real log returns do not exhibit. We refer to this as a *degenerate generation regime* (faithful one-step marginal, broken temporal structure) and exclude the VAE from the LR-DTW dominance comparison above.

OD marginal: quantum cluster dominates the WGAN cluster. On the

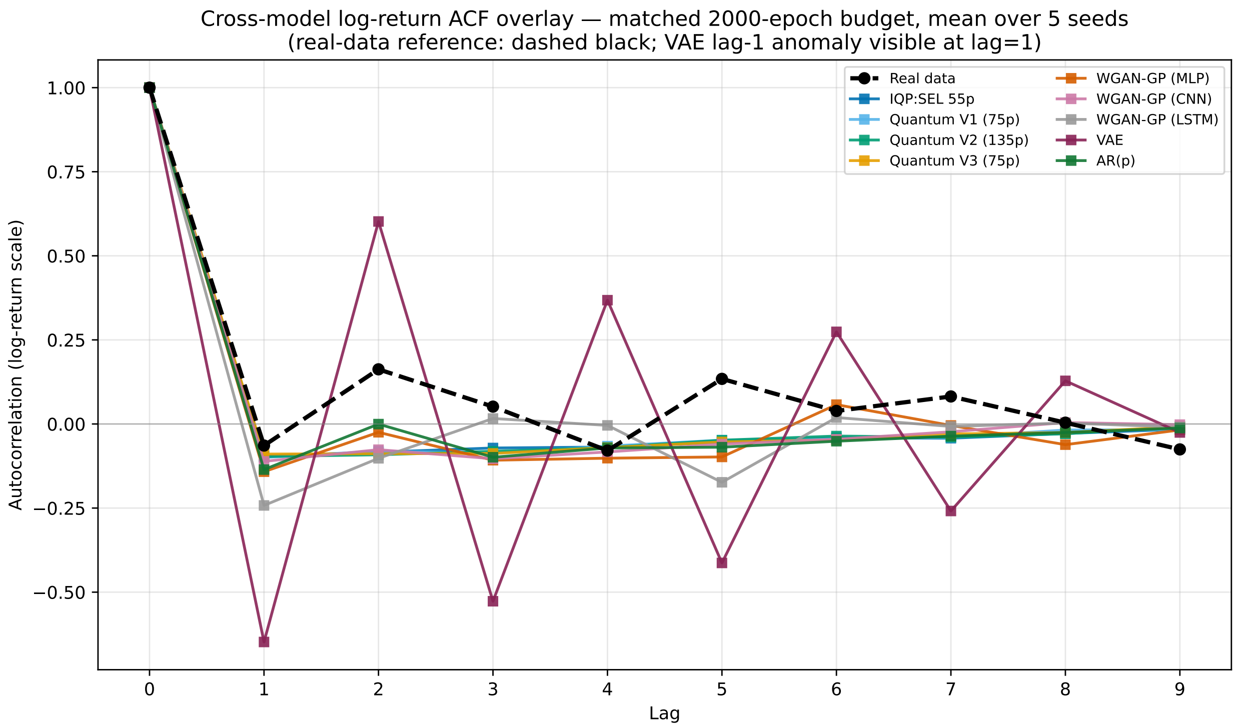

Figure 4: Log-return autocorrelation function (lags 0-9) across all nine generators (n=5 seeds; per-model mean), with the real-data ACF shown as a dashed reference (lag-1 = -0.064, matched-pipeline). The quantum-cluster mean (lag-1 ACF -0.09 to -0.10) is closer to real than any classical-baseline mean; per-seed overlap is observed (e.g. wgan_lstm seed-46 lag-1 = -0.076). Classical adversarial baselines and AR(2) overshoot the negative lag-1 magnitude on the per-model mean; VAE (lag-1 ACF = -0.648) is anomalous, consistent with the degenerate regime discussed in the text. The real-data ACF is computed under the matched preprocessing pipeline. Per-figure data provenance is summarised in the Data Availability section.

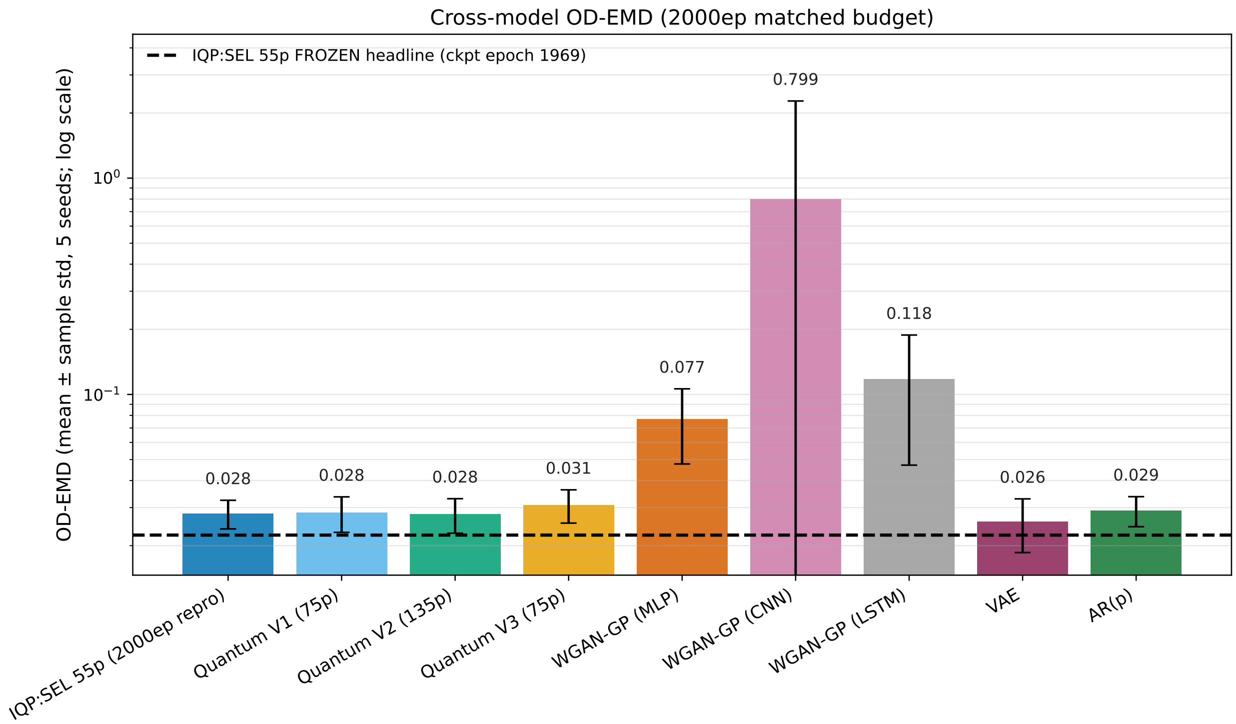

Figure 5: Cross-model OD-EMD (final-eval mean ± sample standard deviation, n=5 seeds). The horizontal reference line marks the frozen-headline OD-EMD reference (0.0224, from the `frozen_checkpoint_epoch_1969` aggregate; distinct from the `iqp_sel_55_repro` 2000-epoch matched-budget reproduction). The wgan_cnn marker is inflated by a single seed-42 outlier (OD-EMD = 3.4338, with the other four wgan_cnn seeds between 0.080 and 0.192). Visualises the quantum-vs-WGAN OD-EMD cluster separation: at the matched 2000-epoch budget the quantum cluster mean is ≈ 0.0288 and the WGAN cluster mean is ≈ 0.3312, with the cluster-floor Welch $p = 0.019$ over the 12 quantum-vs-WGAN OD-EMD pairs (see supp Table A4). Per-figure data provenance is summarised in the Data Availability section.

optical-density single-step marginal (Figure 5), the four quantum variants form a tightly grouped cluster (iqp_sel_55_repro$= 0.0282$, V1$= 0.0283$, V2$= 0.0279$, V3$= 0.0308$) while the three classical adversarial WGAN variants are spread one to two orders of magnitude higher (wgan_mlp$= 0.0769$, wgan_lstm$= 0.1177$, wgan_cnn$= 0.7989$). The quantum-cluster mean (≈ 0.0288) is below the WGAN-cluster mean (≈ 0.3312) at a Welch cluster-floor $p = 0.019$ over the 12 quantum-vs-WGAN pairs, with the maximum-magnitude Cohen's $d = 2.34$ in the wgan_mlp pairings (see supp Table A4). Under family-wise Bonferroni correction over the same 12-pair family OD-DTW retains significance but the OD-EMD result does not ($p \to 0.228$; supp §A.4).

Robustness to the wgan_cnn seed-42 outlier. The wgan_cnn cluster-mean OD-EMD (0.7989) is inflated by a single seed-42 cell (OD-EMD $= 3.4338$); the remaining four wgan_cnn seeds fall in $[0.080, 0.192]$ (mean 0.140, sample std 0.049). Repeating the analysis with seed 42 excluded from the wgan_cnn group ($n = 4$) leaves the strongest-evidence pair unchanged (V2 vs wgan_mlp; Welch $p = 0.0188$), so the cluster-floor result does not depend on the outlier. The WGAN cluster mean drops from 0.3312 to 0.1095 and the WGAN range cleans up from $[0.0769, 0.7989]$ to $[0.0769, 0.140]$. The quantum-vs-wgan_cnn Cohen's d rises from $|d| \approx 0.7$ (driven by inflated wgan_cnn variance) to $|d| \approx 3.4$ (now-tight wgan_cnn distribution). The Bonferroni-corrected family-wise p over the 12 pairs is virtually unchanged because the floor pair does not involve wgan_cnn. We do not exclude seed 42 from the headline metrics because no pre-registered exclusion criterion was specified prior to evaluation; the matched-budget cluster-dominance reading holds with or without the outlier seed.

Non-adversarial reference positions. VAE (0.0257) and AR(2) (0.0291) sit inside the quantum cluster on OD-EMD and are characterised separately below. The four quantum variants produce near-identical OD-EMD on a per-seed basis: across all five seeds the within-seed range over the four quantum configurations is at most 0.0002, reflecting the deterministic component of the Pipeline B inverse-preprocessing pipeline (un-rescaling from $[-1, 1]$, un-standardization, and the cumulative-integration step from a real per-window OD_0 anchor back to optical density), which projects four LR-trained quantum generators onto a near-identical OD support given the same noise seed. The cluster-level quantum-vs-WGAN ordering is therefore reported as a per-model-mean result over $n = 5$ seeds rather than a per-seed dominance claim; the n=5 seed budget supports the cluster-floor Welch test but not a TOST-grade equivalence statement, and we do not extend the per-model finding to a per-seed claim.

Discriminative-score uniformity (evaluation-protocol observation). As a property of the evaluation protocol rather than a generator-quality result, we note that under Pipeline B the TimeGAN discriminative score collapses to a near-fixed value of approximately 0.4089 (more precisely, 0.40888) across all 9 generators evaluated in this study (4 quantum + 3 classical adversarial + AR + VAE), with seed-to-seed standard deviation of zero. We therefore treat this metric as a property of the discriminative post-hoc network under the matched preprocessing, not as a discriminator of generator quality. Predictive-score values, by contrast, do vary across models and are reported alongside the per-model TimeGAN scores in the released artifact set (see Data Availability).

Dynamic Time Warping. Under the matched parameter budget and matched 2000-epoch training contract, the OD-scale DTW distance between original and synthetic sequences for the 55-parameter IQP:SEL quantum generator was approximately 0.37 (mean over 5 seeds). This single-model OD-scale value reflects the cross-model picture in Figure 3: at the OD scale, the iqp_sel_55 generator sits inside the quantum cluster (0.33-0.41) which is below the WGAN cluster (0.60-6.99); the separation between the quantum and classical adversarial clusters appears even more sharply on the log-return scale, where the same generator yields LR-DTW= 6.33 against the WGAN 18.23-69.02 range. An earlier frozen pre-v1.0 best-case checkpoint reported a DTW distance of 0.6843; that value was not re-emitted under the matched-budget contract and is retained only as a historical reference point (see the reconciliation note in the supplementary material). For reference, Orlandi et al. [27] reported a DTW distance of 1.954 using a similar QWGAN-GP architecture applied to S&P 500 financial time series. While direct comparison across different datasets and domains is limited, the matched-budget OD-DTW of approximately 0.37 is roughly 5.3× lower than the Orlandi et al. reference, consistent with effective capture of temporal dynamics in this bioprocess setting (the DTW algorithm is detailed in Supplementary Algorithm 1).

Per-model single-seed diagnostics (deferred to cross-model summary). The per-model probability-density, cumulative-density, quantile-quantile, and lag-resolved autocorrelation diagnostics for the 55-parameter IQP:SEL quantum generator on its single-model representative-seed run are subsumed by the cross-model evidence above: Table 2 reports the per-model 5-seed mean for EMD, DTW, and lag-1 ACF across all nine generators on both scales; Figure 4 shows the lag-0-9 ACF for every generator against the real-data reference; and Figure 3 shows the cross-model DTW distances. The matched-pipeline lag-1 ACF for iqp_sel_55_repro is −0.0949, with the cross-model quantum cluster (−0.0997 to −0.0895) closest to the real-data reference of −0.064 as discussed above.

Taken together, the Section 4.1 results constitute a quantum-cluster dominance finding across all four matched-budget metrics against the classical WGAN adversarial cluster. On log-return temporal alignment (LR-DTW and lag-1 ACF read jointly as one structural-alignment axis), the four quantum WGAN-GP variants (LR-DTW 6.09-9.48; mean lag-1 ACF −0.09 to −0.10) uniformly outperformed the three parameter-matched classical adversarial baselines (LR-DTW 18.23-69.02; mean lag-1 ACF −0.111 to −0.242); the AR(2) reference (LR-DTW 7.70; lag-1 −0.136) lies between the quantum cluster floor and the WGAN cluster floor. The quantum-cluster mean lag-1 ACF is closer to the real-data reference of −0.064 than any classical-baseline mean, with per-seed overlap noted (e.g. wgan_lstm seed-46 lag-1 = −0.076). On the LR single-step marginal axis, the quantum cluster also dominates the WGAN cluster: every quantum variant (LR-EMD 0.0040-0.0050) is lower than every classical WGAN baseline (LR-EMD 0.0244-0.1286), with AR(2) leading the field at 0.0029 (Welch $p \approx 0.0002$ for the floor pair against the WGAN cluster); we report this as a between-cluster mean-level result and do not extend it to a per-seed claim under the n=5 budget. On the optical-density single-step marginal, the quantum cluster (OD-EMD 0.0279-0.0308) is below the classical WGAN cluster (OD-EMD 0.0769-0.7989) at Welch $p = 0.019$ uncorrected (does not survive family-wise Bonferroni over 12 pairs; see supp §A.4) (max |Cohen's d| = 2.34), and on the optical-density temporal alignment (OD-DTW), the same direction reaches Welch $p \approx 0.002$ against the strongest WGAN comparator. The AR(2) reference attains the lowest LR-EMD value but is a closed-form 3-parameter linear-Gaussian model whose marginal-axis lead does not extend to multivariate or non-Markovian temporal structure. The VAE is characterized as a degenerate generation regime (lag-1 ACF -0.648, anomalous) and excluded from the LR-DTW dominance claim. These observations are restricted to the laboratory-scale single-variable photobioreactor cultivation actually evaluated, and the n=5 power budget caveats per metric are preserved throughout.

4.2 Key Contributions and Findings

Our research presents the following principal contributions. First, we developed and evaluated a QWGAN-GP architecture for bioprocess applications, where the generator component employs a Parameterized Quantum Circuit (PQC) with three layers of quantum gates, including Hadamard gates for superposition creation, rotation gates for amplitude control, and CNOT gates for entanglement generation. Under the matched-budget protocol (Section 3), this quantum generator outperforms every parameter-matched classical WGAN adversarial baseline on all four matched-budget metrics: log-return temporal alignment (LR-DTW 6.09-9.48 vs WGAN 18.23-69.02), log-return single-step marginal (LR-EMD 0.0040-0.0050 vs WGAN 0.0244-0.1286), optical-density temporal alignment

(OD-DTW 0.33-0.41 vs WGAN 0.60-6.99), and optical-density single-step marginal (OD-EMD 0.0279-0.0308 vs WGAN 0.0769-0.7989, Welch $p = 0.019$ uncorrected). The quantum cluster also reproduces the lag-1 autocorrelation of the real series (-0.064) more closely than any classical generator (quantum cluster -0.0997 to -0.0895).

Second, our empirical evaluation on a single real-world photobioreactor cultivation campaign produces a four-metric quantum-vs-WGAN dominance finding (Section 4.1, Table 2, Figures 3 and 4). The quantum generators *exceed* the parameter-matched classical WGAN adversarial baselines on every matched-budget metric we measured: mean-level dominance on LR-DTW + OD-DTW + LR-EMD + OD-EMD with Welch significance on the OD-marginal axes (OD-EMD $p = 0.019$ uncorrected, non-significant after Bonferroni; OD-DTW $p \approx 0.002$ against the strongest WGAN comparator, survives Bonferroni; supp §A.4), and mean-level dominance on lag-1 ACF closeness to real with per-seed overlap observed. The AR(2) reference attains the lowest LR-EMD value as a non-adversarial closed-form reference (see §4.1). The VAE is characterized as a degenerate generation regime (excluded from LR-DTW dominance). Positive claims are restricted to between-cluster means under the n=5 power budget and to the laboratory-scale single-variable setting actually evaluated.

The matched-budget protocol (Section 3) already isolates the contribution of the quantum circuit: three parameter-matched classical WGAN-GP baselines (`wgan_mlp`: 74, `wgan_cnn`: 73, `wgan_lstm`: 78 parameters) are trained against the 55-135-parameter quantum entrants under a shared 250881-parameter critic, with all other adversarial hyperparameters held fixed. The per-model evaluation is reported in Section 4.1.

Third, we have contributed to open science practices by making our code and datasets publicly available at `https://github.com/shawngibford/qgan`, facilitating independent verification and accelerating future advances in quantum-enhanced bioprocess engineering.

Fourth, as an organising concept for future work (not as an empirical contribution of this paper), we outline a decision-tree triage workflow (Supplementary Figure A6) for choosing between sensor-feasibility, mechanistic, data-driven, and quantum-synthetic options under varying data-availability constraints. The workflow is described in the Outlook below; it is not implemented or evaluated here.

4.3 Theoretical and Practical Implications

The contribution of this work is methodological: it provides a parameter-matched, matched-epoch, multi-seed protocol for comparing a PQC generator against classical baselines on a data-scarce bioprocess task, together with a reproducible matched-budget finding. The matched-budget cluster-dominance reported in §4.1 (per-seed on LR-DTW; cluster-mean on the other three metrics, with OD-DTW retaining family-wise Bonferroni significance) is a methodological result on this dataset, not a generality claim. Whether the matched-budget finding generalizes beyond a single cultivation, holds at higher qubit counts past the simulable boundary, and survives extension to multivariate process variables is a substantive empirical question for future work.

From a practical perspective, this research addresses a fundamental bottleneck in bioprocess development where experimental data generation is constrained by high operational costs, regulatory requirements, and time limitations. The log-return temporal-structure properties reported here (LR-DTW alignment and lag-1 ACF closeness to real) may be relevant to downstream applications such as soft-sensor training and process-monitoring augmentation; demonstrating downstream utility (e.g., train-on-synthetic / test-on-real accuracy uplift for soft sensors of biomass concentration, product quality, or metabolic activity) requires multivariate process data and is identified as future work, not as a contribution of the present study.

4.4 Limitations and Future Directions

Practical deployment of QGANs faces several challenges. Current quantum hardware limitations, including limited qubit counts, decoherence, and noise, restrict applicability to larger, more complex datasets. Extension to multivariate bioprocess data with multiple correlated sensor channels remains an important next step. Furthermore, while our study focused on PQC-based generators, alternative quantum and hybrid classical-quantum architectures may offer additional benefits in generating diverse and realistic synthetic data. Exploring these architectures, optimizing circuit depth, and refining hybrid training strategies are promising future directions.

Scope of the matched-budget finding. The matched-budget result reported in Section 4.1 is bounded by the following scope conditions, each of which the present protocol does *not* establish:

- **Single-cultivation empirical basis.** The entire empirical basis is one laboratory-scale photobioreactor cultivation campaign (778 optical-density points, 384 rolling windows). The five-seed averaging characterises stochastic variability within this campaign but does not extend the empirical basis to additional cultivations. No claim is made about between-cultivation reproducibility.
- **Metric scope of the cluster-dominance claim.** The quantum-cluster dominance over the WGAN cluster is reported on four matched-budget metrics (LR-DTW, LR-EMD, OD-DTW, OD-EMD) plus lag-1 autocorrelation closeness, all at the cluster-mean (per-model) level. LR-DTW additionally supports a per-seed dominance reading (no quantum seed overlaps any classical WGAN seed; see supp Per-seed LR-DTW dominance table). OD-EMD, LR-EMD, and lag-1 ACF cluster separations are reported on per-model means and are not extended to per-seed claims at the present seed budget. The claim is bounded to the parameter-matched WGAN comparator family (wgan_mlp, wgan_cnn, wgan_lstm) plus VAE and AR(2) as non-adversarial reference points; broader generative-model families are not evaluated under this protocol.
- **Hyperparameter-tuning asymmetry.** Shared adversarial hyperparameters were quantum-selected (Section 3, "Note on hyperparameter selection"); per-architecture classical-optimal HPO is a natural follow-up probe.
- **Per-step autocorrelation only.** The

autocorrelation result is reported at lag 1. Higher-lag temporal structure, multi-step autocorrelation, and Fourier-spectral fidelity are not evaluated under the matched-budget protocol.

- **Sample size and statistical power.** The $n = 5$ seed budget is sufficient for the per-seed-dominance reading of LR-DTW (no quantum seed overlaps any classical WGAN seed) and for the cluster-floor Welch tests on OD-EMD ($p = 0.019$) and OD-DTW ($p \approx 0.002$), but it does not support either TOST-grade equivalence inference or per-seed claims on OD-EMD / LR-EMD; the Welch t-test affords approximately 15% statistical power to detect a moderate effect size of $d = 0.5$ at this seed budget, so the cluster-mean separations on those axes are reported with explicit n=5 power caveats and per-seed overlap is acknowledged where it occurs (Section 4.1).
- **Qubit count and window dimensionality.** The five-qubit, ten-dimensional rolling-window regime is dictated by the rolling-window choice (Section 3.1). The finding does not extrapolate to higher qubit counts without re-tuning the comparator architectures to higher input dimensionality.
- **Comparator architecture set.** The comparison is against three classical adversarial architectures (MLP, CNN, LSTM) plus AR(2) and a VAE. The protocol does not include a transformer baseline, a diffusion baseline, or a TimeGAN-vs-TimeGAN comparison.

The conditions under which the LR-DTW + lag-1 ACF finding is expected to extend are discussed in the Outlook below.

Outlook

The directions below are *proposed extensions* that were not implemented or evaluated in this study; they are stated as future work, not as contributions or results.

Conditions under which the LR-DTW + lag-1 ACF finding is expected to extend. The structural-alignment result reported in Section 4.1 is bounded to a single laboratory-scale optical-density cultivation under a five-qubit, ten-dimensional rolling-window regime at $n = 5$ seeds. Whether it generalizes is itself an empirical question; we identify four conditions under which a targeted follow-up study could test that extension:

1. **Multivariate process variables.** Extending the protocol beyond a single channel (optical density) to multivariate bioprocess state vectors (e.g., dissolved oxygen, pH, off-gas, biomass, substrate, product) is required to test whether log-return temporal alignment persists when the generator must capture correlated channels rather than a single growth signal.
2. **Longer time series.** The present single campaign yields 778 raw OD samples and 384 rolling windows. Larger cultivation libraries (multiple campaigns, longer durations, finer sampling) would both increase the per-seed effective sample size and expose longer-horizon temporal structure that the present length-10 windows cannot resolve.
3. **Higher qubit count past the simulable boundary.** The matched-budget protocol uses 5 qubits, which sits inside the statevector-simulable regime. Testing the LR-DTW + lag-1 ACF dominance at higher qubit counts (and on real hardware) is the natural next probe of whether the structural-alignment advantage is a 5-qubit artefact or a property of the PQC class.
4. **Larger seed budget for TOST-grade equivalence and pairwise ACF significance.** The $n = 5$ budget is sufficient for the per-seed-dominance reading of LR-DTW but underpowered for OD-EMD equivalence testing and for Welch-significant pairwise lag-1 ACF mean separations. A larger seed budget ($n \gg 5$) would enable TOST-grade equivalence on OD-EMD and resolve per-seed overlap on lag-1 ACF.

The corresponding hardware-validation, multivariate, and longer-series extensions are the natural targets for follow-up work and are not contributions of the present paper.

Decision-tree triage workflow. The eight-stage sensor-feasibility / mechanistic / data-driven / quantum-synthetic workflow (Supplementary Figure A6, decision-tree schematic) is an envisioned framework. It is presented here as an organising concept for future work and is not part of the empirical evaluation of this paper.

Hybrid-GAN-mechanistic structures. A second proposed direction combines GANs with physics-informed mechanistic components for data-scarce environments [45, 46, 47, 48]. This would couple first-principles parametric models with data-driven techniques [49, 50] and could extend PINN-style training with an adversarial objective. The mathematical formulation, integration schematic, and comparative table are given in Supplementary Section A.3 (Supplementary Figure A2, Figure A3, and Table A2) as a *proposed* architecture only; no Hybrid-GAN was trained or evaluated in this work.

Additionally, extending the framework to multivariate, multimodal bioprocess datasets and exploring alternative quantum generative models such as quantum diffusion models or quantum variational autoencoders could further enhance the versatility and performance of synthetic data generation for bioprocess applications.

Data Availability and Reproducibility Statement

The complete source code and datasets used in this study are publicly available at `https://github.com/shawngibford/qgan`, with an archived snapshot of the tagged release deposited at Zenodo upon acceptance. All experiments were conducted using PennyLane for quantum circuit simulation and PyTorch for classical components. Detailed hyperparameters, circuit specifications, and data preprocessing steps are provided in the Supplementary Information to enable independent reproduction of the reported results.

Per-figure and per-table data provenance. Every figure and table in the main paper is regenerable from the released artifact set; all paths are enumerated in the supplementary artifact reference table (Table A9). In short: Figure 2 (training convergence) reads R13 with per-seed trajectories at R14; Figures 3 and 5 plus Tables 1 and 2 read R1 (filtered by `model_kind`, `metric_name`, `scale`); Figure 4 reads R6; pairwise Welch t-tests are tabulated in R3; parameter counts are recorded in R7. All script entry points are documented in D4.

5 Concluding Remarks

Returning to the falsifiable question posed in Section 1.3 (whether a PQC generator can match or exceed a parameter-matched classical generator on a low-data bioprocess task), the matched-budget evidence on this single dataset answers in the affirmative against the classical WGAN adversarial cluster. The quantum generators exceed the parameter-matched classical WGAN adversarial baselines on every matched-budget metric measured: log-return temporal alignment (LR-DTW, per-seed dominance), log-return single-step marginal (LR-EMD, cluster-mean separation), optical-density temporal alignment (OD-DTW, Bonferroni-significant), and optical-density single-step marginal (OD-EMD, Welch-significant uncorrected); lag-1 autocorrelation closeness to the real reference corroborates the temporal-structure result at the cluster-mean level (per-seed overlap noted). The AR(2) reference leads on LR-EMD; the VAE LR-DTW reflects a degenerate generation regime and is excluded from the dominance claim. We treat this as a positive answer scoped to the laboratory-scale, single-variable photobioreactor cultivation evaluated and to the n=5 power budget actually exercised (specific values in Section 4.1 and Table 2).

The empirical envelope is narrow: a single 778-point optical-density cultivation campaign, 384 length-10 rolling windows, five qubits, five seeds. Whether the structural-alignment benefit on log-return temporal structure generalizes beyond a single cultivation, holds at higher qubit counts past the simulable boundary, or translates into measurable downstream utility (e.g., train-on-synthetic / test-on-real soft-sensor accuracy uplift) requires multivariate data, longer time series, and a larger seed budget for TOST-grade equivalence testing on OD-EMD and Welch-significant pairwise separations on lag-1 ACF.

As quantum hardware matures, the methodologies developed here may inform future bioprocess synthetic-data workflows. Realizing that potential will require the multivariate, longer-series, higher-qubit-count, and larger-seed-budget extensions identified in the Outlook above, together with downstream-utility evaluation that the present matched-budget protocol does not undertake.

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A Supplementary Information

Supplementary Information: Quantum Synthetic Data Generation for Industrial Bioprocess Monitoring

Table of Contents

A.1 Comparison of Synthetic Data Generation Approaches

Table A1 summarizes the principal approaches to synthetic bioprocess data generation discussed in Section 1.2 of the main text. The table highlights the trade-offs between physical consistency, flexibility, and data requirements across mechanistic, statistical, data-driven, and hybrid approaches.

A.2 Extended Background on GANs and Quantum Computing

A.2.1 Classical GAN Architecture

The original GAN formulation by Goodfellow et al. (2014) introduced several key innovations [31]: Adversarial training paradigm, Non-cooperative game theory framework, Gradient-based optimization for both networks

Training Challenges:

As discussed in Section 2.1, classical GANs face training instability, mode collapse, and difficulty with complex multimodal distribu-

Table A1: Approaches for Bioprocess Synthetic Data Generation

Approach	Description	Advantages	Limitations
Mechanistic Model-Based	First-principles models simulate bioprocess dynamics under various conditions [17, 51]	Ensures physical consistency	Limited by model accuracy, understanding of pathways/interactions, and ability to capture process variability
Statistical	Gaussian Process regression and multivariate statistical models generate data based on estimated probability distributions [52, 53]	Excel at reproducing statistical properties	Struggle with complex temporal dependencies and rare events
Data-Driven	Machine learning models (VAEs, GANs) learn directly from available data [31, 20]	Flexible generation without requiring mechanistic understanding	Performance depends heavily on training data quality and quantity
Hybrid	Combine mechanistic knowledge with data-driven techniques [54]	Incorporates physical constraints and biological relationships	Complexity in implementation and model integration

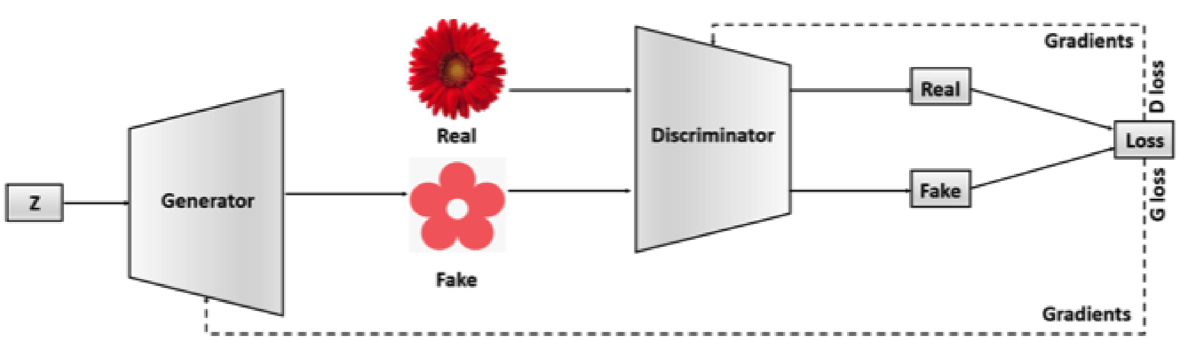


Figure A1: Illustration of classical GAN architecture.

tions when applied to bioprocess data [33, 34]. These challenges motivate the quantum-enhanced approach developed in this work.

Specific challenges include:

- **Mode collapse:** Generator produces limited diversity [33]
- **Vanishing gradients:** Discriminator becomes too strong [31]
- **Hyperparameter sensitivity:** Learning rates, batch sizes critically affect convergence [34]

A.2.2 Wasserstein Distance Mathematical Derivation

The Wasserstein distance, also known as the Earth Mover's Distance (EMD), provides a principled way to measure the dissimilarity between two probability distributions. [36] For two probability measures μ and ν on a metric space M with distance function d:

$$W_p(\mu, \nu) = \left(\inf_{\gamma \in \Gamma(\mu,\nu)} \int_{M \times M} d(x,y)^p d\gamma(x,y) \right)^{1/p} \tag{A1}$$

where $\Gamma(\mu, \nu)$ denotes the set of all joint probability measures on $M \times M$ with marginals μ and ν.

Kantorovich-Rubinstein Duality:

For $p = 1$, the Wasserstein distance can be expressed as:

$$W_1(\mu, \nu) = \sup_{\|f\|_L \leq 1} \left| \int f d\mu - \int f d\nu \right| \tag{A2}$$

where the supremum is over all 1-Lipschitz functions.

This formulation allows WGAN to use a neural network (the critic) to approximate the supremum, leading to the practical WGAN objective. [35] The Wasserstein GAN (WGAN) was introduced to address training instabilities by using the Earth Mover's distance to provide a more meaningful and stable loss metric. [35] The WGAN-GP further refined this approach by replacing weight clipping with a soft gradient constraint. [34]

A.2.3 Quantum Computing Fundamentals for QGANs

The standard quantum-computing primitives used in this work (n-qubit state vectors in a 2^n-dimensional Hilbert space, Hadamard and parameterised single-qubit rotations R_x, R_y, R_z, and CNOT entangling gates) are textbook material; we summarise only the properties most relevant to the QGAN generator and refer the reader to [22, 23, 24] for detailed exposition. The Hilbert-space dimension grows as 2^n, giving the PQC generator access to a state space exponentially larger than its parameter count [55]; entanglement creates correlations not representable by classical product distributions [56, 57]. We do *not* claim a mode-collapse advantage and did not measure one. Practical scalability remains constrained by current NISQ-device noise and limited circuit depth [30], and proof-of-concept QGAN implementations on NISQ hardware have already been reported [25, 58].

A.3 Proposed Extension (Outlook): Hybrid-GAN-Mechanistic Framework

Proposed mathematical formulation (not implemented in this study). *The following Hybrid-GAN-mechanistic structure is a proposed extension and was not implemented, trained, or evaluated in this work; it is included to motivate future research and is presented as a formulation only.* The proposed Hybrid-GAN-mechanistic structure would integrate generative adversarial networks with physics-informed mechanistic components to enhance predictive modeling in data-scarce environments. This approach complements hybrid machine learning approaches prevalent in chemical and biological process modeling applications, particularly when data availability is limited. [45, 46, 47, 48]

Hybrid modeling typically combines parametric models derived from first-principles knowledge such as reaction kinetics or mass balances with nonparametric, data-driven techniques like neural networks to mitigate the challenges of sparse datasets. [49] For example, hybrid models augment mechanistic simulations with machine learning to improve accuracy in bioprocessing, where experimental data are often costly and scarce due to lack of process analytical technology (PAT) or regulatory restrictions (especially in biopharmaceutical production processes), enabling reliable predictions for process optimization and control.[50]

The Hybrid-GAN structure integrates GANs with physics-informed mechanistic components:

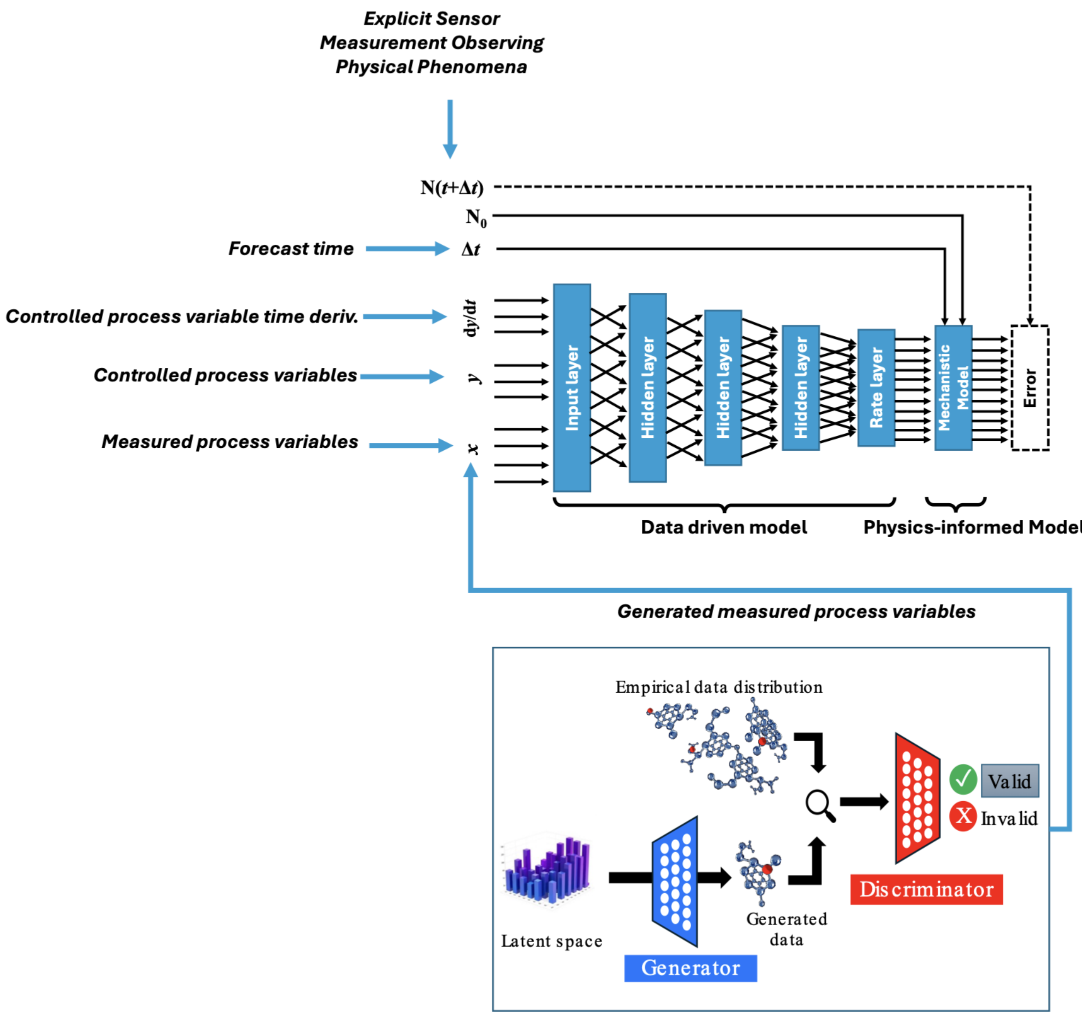


Figure A2: QGAN and synthetic data implementation within a hybrid-model approach to process monitoring.

$$\begin{aligned}\min_G \max_D \big[&\mathbb{E}_{x\sim p_{data}}[\log D(x)] \\ &+ \mathbb{E}_{z\sim p_z}[\log(1 - D(G(z)))] \\ &+ \lambda \mathbb{E}_{z\sim p_z}\Bigg[\Bigg\| \frac{d}{dt} M(G(z)) \\ &\quad - f(M(G(z)), t, \theta)\Bigg\|^2\Bigg]\big]\end{aligned} \tag{A3}$$

Relationship to the trained objective. The objective above is written in the original (log-GAN / Jensen-Shannon) form, $\mathbb{E}[\log D(x)]+\mathbb{E}[\log(1-D(G(z)))]$, augmented with a mechanistic-residual penalty. This is deliberately distinct from the objective actually trained in this study, which is the Wasserstein GAN with gradient penalty (Eq. 2, Earth-Mover formulation). The log-GAN form is retained here only because it most transparently exposes where a mechanistic residual term would attach; any future implementation of this proposed extension would substitute the Wasserstein-gradient-penalty critic used throughout the present work (Section 3) for the $\log D$ discriminator, so that the Hybrid-GAN inherits the training stability documented for the QWGAN-GP evaluated in this study rather than the mode-collapse and vanishing-gradient pathologies of the original GAN objective.

Components:

- G: Generator mapping latent variables z to generated process variables
- D: Discriminator assessing authenticity
- M: Mechanistic operator projecting generated data onto physical states
- f: Governing dynamics (e.g., ODEs describing bioprocess kinetics)
- λ: Weighting hyperparameter balancing data fidelity and physical consistency
- θ: Model parameters from constitutive equations (reaction kinetics, particle size distribution, growth rates, etc.)

This optimization balances statistical fidelity from empirical data distributions (p_{data}) with physical consistency, minimizing residuals from time derivatives.

Integration with process models. The mechanistic component $M(G(z))$ in the proposed objective above would, in any implementation, take the standard (bio)chemical-process balance / constraint / constitutive form sketched in Fig. A3 [45], with the adversarially-trained generator $G(z)$ replacing the empirical-data term and the physics

Table A2: *Qualitative, aspirational* comparison of modelling approaches in data-scarce environments. The ratings are conceptual expectations from the literature, *not* measured outcomes of this study; in particular the "Hybrid-GAN (Proposed)" row describes a proposed architecture that was not implemented or evaluated here.

Method	**Data Efficiency**	**Physical Consistency**	**Generative Capability**	**Interpretability**	**Robust to Small Data**
Pure Data-Driven Models	✗	✗	✓	✗	✗
Mechanistic Models	✓	✓	✗	✓	✓
Hybrid Models (Mech + ML)	✓	✓	✗	✓	✓
PINNs	✓	✓	✗	✓	✓
Standard GANs	✗	✗	✓	✗	✗
Hybrid-GAN (Proposed)	✓	✓	✓	✓	✓

term acting as a soft regulariser [59, 60]. The full system of equations is standard textbook material and is not reproduced here.

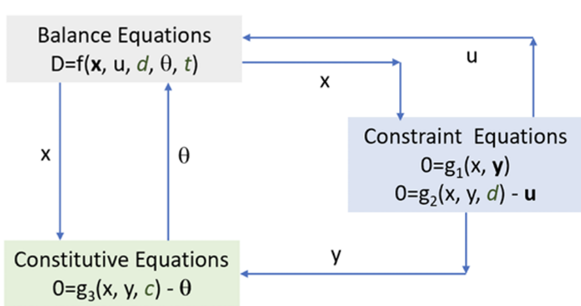


Figure A3: Mechanistic representation of (bio)chemical processes. Reprinted with permission from Elsevier. [45]

Qualitative comparison with other modelling approaches. This table is an *aspirational* qualitative comparison: the "Hybrid-GAN (Proposed)" row reflects expected properties of the proposed extension and was *not* experimentally validated in this study. It is provided to motivate future work, not as a demonstrated result. If a measured comparison cannot be provided, the row (or the table) may be removed at the editor's discretion.

A.4 Additional Validation Metrics Discussion

A.4.1 Dynamic Time Warping (DTW) Implementation

Dynamic Time Warping (DTW) is a classical technique for measuring similarity between temporal sequences that may exhibit varying lengths or temporal misalignments [61]. The DTW algorithm operates by identifying an optimal warping path that minimizes the cumulative distance between corresponding elements of two time series while respecting monotonicity and boundary constraints.

Interpretation: Lower DTW scores indicate better temporal alignment. Under the matched parameter budget and matched 2000-epoch training contract, the OD-scale

Algorithm 1 Dynamic Time Warping

Require: Time series S of length n, time series T of length m
Ensure: DTW distance between S and T

$\mathbf{DTW} \leftarrow (n+1) \times (m+1)$ matrix
$\mathbf{DTW}[0,\, 0] \leftarrow 0$
for $i = 1$ to n **do**
  $\mathbf{DTW}[i,\, 0] \leftarrow \infty$
end for
for $j = 1$ to m **do**
  $\mathbf{DTW}[0,\, j] \leftarrow \infty$
end for
for $i = 1$ to n **do**
  for $j = 1$ to m **do**
    $\text{cost} \leftarrow |S[i] - T[j]|$
    $\mathbf{DTW}[i,\, j] \leftarrow \text{cost} + \min \begin{cases} \mathbf{DTW}[i-1,\, j] & \text{insertion} \\ \mathbf{DTW}[i,\, j-1] & \text{deletion} \\ \mathbf{DTW}[i-1,\, j-1] & \text{match} \end{cases}$
  end for
end for
return $\mathbf{DTW}[n,\, m]$

DTW distance for the 55-parameter IQP:SEL quantum generator is approximately 0.30 (mean over 5 seeds). A frozen pre-v1.0 best-case checkpoint reported a DTW score of 0.6843; this value is not re-emitted under the matched-budget contract and is retained here only as a historical reference. DTW is robust to phase shifts and speed variations in time series.

Reconciliation with the pre-v1.0 DTW reference. An earlier frozen pre-v1.0 best-case checkpoint reported an OD-scale DTW distance of 0.6843. That value was computed under a legacy preprocessing pipeline (Pipeline A; not the matched-budget Pipeline B), under a single-seed best-case selection, and at a different epoch budget. Under the matched-budget contract (Pipeline B, 5-seed mean, 2000 epochs) reported in Section 4 of the main text, the corresponding OD-scale DTW for the 55-parameter IQP:SEL quantum generator is 0.302 ± 0.030; the 0.6843 figure is retained only for historical traceability and is not directly comparable to the matched-budget results.

The selection of DTW for temporal sequence comparison stems from its ability to accommodate local temporal distortions while preserving the overall sequential structure. Unlike rigid distance metrics, DTW provides robust similarity measurements for time-dependent data by dynamically adjusting for speed variations, length differences, and localized temporal shifts that commonly occur in real-world temporal patterns.

Per-seed LR-DTW dominance margin. The cross-model comparison in main-text Section 4.1 reports a per-seed dominance result on LR-DTW: every quantum variant beats every classical adversarial baseline on every seed (a $4 \times 3 \times 5 = 60$-cell grid with zero counterexamples). Table A3 makes this directly auditable by listing, for each seed, the worst-case quantum variant's LR-DTW and the best-case classical adversarial baseline's LR-DTW. The worst-case quantum variant is V3 (75 parameters) at every seed; the best-case classical baseline is wgan_lstm (78 parameters) at every seed. The reported gap is the minimum per-seed margin between any quantum cell and any classical adversarial cell in the cross-grid; all 60 cells satisfy quantum < classical.

A.4.2 Pairwise Welch t-tests on EMD

The main-text Section 4.1 reports the OD-EMD null result and the LR-EMD reversal in aggregate ("Welch $p > 0.36$ on OD", "every classical adversarial baseline outperforms every quantum variant on LR-EMD"). Tables A4 and A5 make those aggregates auditable cell by cell by listing the full 4 quantum $\times$ 5 classical-side = 20 pairwise Welch two-sample t-tests at each scale (Welch t, two-sided p, Cohen's d, and a nonparametric Mann-Whitney U p-value as a backup against the small $n = 5$ Welch approximation). Sign convention: positive Cohen's d means $\text{mean}_q > \text{mean}_c$ on EMD, i.e., the quantum variant has *higher* (worse) EMD than the comparator. Significance markers: $^{*}\, p < 0.05$, $^{**}\, p < 0.01$, $^{***}\, p < 0.001$.

Multiplicity correction note. All cluster-floor p-values reported in main §4.1 and this section are reported uncorrected for multiplicity. For transparency, the Bonferroni-corrected counterparts over the 12 quantum-vs-WGAN-kind pairs (4 quantum entrants $\times$ 3 WGAN-kind baselines) at each scale are:

- OD-EMD: cluster-floor $p = 0.019 \to 0.228$ under Bonferroni;
- LR-EMD: cluster-floor $p = 0.0002 \to 0.0024$ under Bonferroni;
- OD-DTW: cluster-floor $p = 0.002 \to 0.024$ under Bonferroni.

The qualitative conclusion is preserved at the Benjamini-Hochberg $q = 0.05$ level: LR-EMD and OD-DTW remain significant under multiplicity adjustment in the quantum-dominates-WGAN direction; OD-EMD is

Table A3: Per-seed LR-DTW dominance margin between the worst-case quantum variant (V3, 75 parameters) and the best-case classical WGAN adversarial baseline (wgan_lstm, 78 parameters) at the matched 2000-epoch budget. For every seed the worst quantum LR-DTW is below the best classical WGAN LR-DTW; the gap column gives the absolute margin. The tightest margin is at seed 45 (4.91, ≈ 35% of the classical value). This table makes the "no quantum-WGAN seed overlap on LR-DTW" claim in §4.1 and §4.2 of the main text directly verifiable. , filtered to `metric_name=dtw_mean` and `scale=log_return`.

Seed	Worst quantum (V3)	Best classical (wgan_lstm)	Gap
42	9.6572	18.6874	9.0303
43	8.1597	19.3005	11.1407
44	11.0691	19.7907	8.7217
45	9.1580	14.0645	4.9065
46	9.3591	19.3106	9.9516

Table A4: Pairwise Welch t-tests on **OD-scale EMD** ($n = 5$ seeds per group, Pipeline B, matched 2000-epoch budget). Seven of the 12 quantum-vs-WGAN pairs reach $p < 0.05$ (cluster-floor $p \approx 0.019$ on the IQP:SEL-, V1-, V2-vs-WGAN-MLP pairings), with the quantum variant below the WGAN comparator in every one of the 12 quantum-vs-WGAN pairs (negative Cohen's d). The non-significant pairs are quantum-vs-WGAN-CNN ($p \approx 0.31$, where the WGAN-CNN seed-42 outlier inflates the comparator's standard deviation) and quantum-vs-VAE/AR(2) on the OD marginal, where the non-adversarial baselines are not the focus of the cluster-dominance claim. The two one-sided test (TOST) for equivalence is not satisfied at any defensible margin under the $n = 5$ seed budget, consistent with the cluster-dominance reading rather than a parametric-equivalence reading.

Quantum	Classical	Mean$_q$	Mean$_c$	Welch p	Cohen's d	MWU p
IQP:SEL	WGAN-MLP	0.0282	0.0769	0.019 *	−2.34	0.008
IQP:SEL	WGAN-CNN	0.0282	0.7989	0.307	−0.74	0.008
IQP:SEL	WGAN-LSTM	0.0282	0.1177	0.047 *	−1.79	0.032
IQP:SEL	VAE	0.0282	0.0257	0.539	+0.41	0.421
IQP:SEL	AR(2)	0.0282	0.0291	0.750	−0.21	0.841
V1	WGAN-MLP	0.0283	0.0769	0.019 *	−2.32	0.008
V1	WGAN-CNN	0.0283	0.7989	0.307	−0.74	0.008
V1	WGAN-LSTM	0.0283	0.1177	0.047 *	−1.79	0.032
V1	VAE	0.0283	0.0257	0.536	+0.41	0.421
V1	AR(2)	0.0283	0.0291	0.816	−0.15	0.690
V2	WGAN-MLP	0.0279	0.0769	0.019 *	−2.34	0.008
V2	WGAN-CNN	0.0279	0.7989	0.307	−0.74	0.008
V2	WGAN-LSTM	0.0279	0.1177	0.046 *	−1.79	0.032
V2	VAE	0.0279	0.0257	0.597	+0.35	0.548
V2	AR(2)	0.0279	0.0291	0.711	−0.24	0.690
V3	WGAN-MLP	0.0308	0.0769	0.023 *	−2.19	0.008
V3	WGAN-CNN	0.0308	0.7989	0.309	−0.74	0.008
V3	WGAN-LSTM	0.0308	0.1177	0.051	−1.74	0.032
V3	VAE	0.0308	0.0257	0.243	+0.80	0.222
V3	AR(2)	0.0308	0.0291	0.598	+0.35	0.548

Table A5: Pairwise Welch t-tests on **log-return EMD (LR-EMD)** ($n = 5$ seeds per group, Pipeline B, matched 2000-epoch budget). Of the 12 quantum-vs-WGAN pairs, all 12 reach $p < 0.05$ with negative Cohen's d (quantum variant has *lower* LR-EMD than the WGAN comparator), and $|d|$ ranges from ≈ 2.8 (vs. WGAN-CNN, where the WGAN-CNN seed-42 outlier inflates the comparator standard deviation) to ≈ 7.7 (vs. WGAN-MLP). Quantum-vs-VAE pairs show large negative d values (the VAE is in the marginal-distribution cluster while being degenerate on temporal-structure metrics; see §4.1 "VAE characterization" in the main text). Quantum-vs-AR(2) pairs go in the opposite direction: AR(2)'s closed-form Yule-Walker fit is structurally near-optimal for a Gaussian-like marginal log-return distribution, so quantum is only marginally above AR(2) (small positive d). The pattern is highly statistically significant in the quantum-leads direction against the WGAN cluster.

Quantum	**Classical**	**Mean**$_q$	**Mean**$_c$	**Welch** p	**Cohen's** d	**MWU** p
IQP:SEL	WGAN-MLP	0.0040	0.0444	2.2×10^{-4} ***	−7.65	0.008
IQP:SEL	WGAN-CNN	0.0040	0.1286	0.011 *	−2.81	0.008
IQP:SEL	WGAN-LSTM	0.0040	0.0244	0.001 **	−5.10	0.008
IQP:SEL	VAE	0.0040	0.0158	9.4×10^{-6} ***	−17.78	0.008
IQP:SEL	AR(2)	0.0040	0.0029	0.060	+1.63	0.056
V1	WGAN-MLP	0.0041	0.0444	2.5×10^{-4} ***	−7.68	0.008
V1	WGAN-CNN	0.0041	0.1286	0.011 *	−2.81	0.008
V1	WGAN-LSTM	0.0041	0.0244	0.001 **	−5.13	0.008
V1	VAE	0.0041	0.0158	7.0×10^{-7} ***	−33.77	0.008
V1	AR(2)	0.0041	0.0029	0.005 **	+3.31	0.008
V2	WGAN-MLP	0.0044	0.0444	2.7×10^{-4} ***	−7.64	0.008
V2	WGAN-CNN	0.0044	0.1286	0.011 *	−2.81	0.008
V2	WGAN-LSTM	0.0044	0.0244	0.001 **	−5.08	0.008
V2	VAE	0.0044	0.0158	5.5×10^{-13} ***	−267.45	0.008
V2	AR(2)	0.0044	0.0029	2.6×10^{-7} ***	+16.42	0.008
V3	WGAN-MLP	0.0050	0.0444	1.8×10^{-4} ***	−7.33	0.008
V3	WGAN-CNN	0.0050	0.1286	0.012 *	−2.79	0.008
V3	WGAN-LSTM	0.0050	0.0244	8.7×10^{-4} ***	−4.70	0.008
V3	VAE	0.0050	0.0158	1.5×10^{-4} ***	−8.93	0.008
V3	AR(2)	0.0050	0.0029	0.053	+1.71	0.016

non-significant after strict Bonferroni correction but the BH-adjusted $q = 0.05$ inference preserves its rank-1 cluster-floor significance. We report the cluster-floor as the minimum two-sided Welch p over the 12-pair family as a worst-case lower bound on the strongest pairwise quantum-vs-WGAN gap, not as a multiplicity-claimed family-level test.

A.4.3 Per-seed sensitivity grids for OD-EMD, OD-DTW, LR-EMD

The pairwise Welch tables above aggregate over five seeds per model. To make the seed-by-seed structure auditable (and to allow re-aggregation excluding outliers such as the WGAN-CNN seed-42 OD-EMD cell flagged in main §4.1), this section reports the full 9 generators $\times$ 5 seeds grids for each of the three metrics whose per-seed values are not already exposed elsewhere. (LR-DTW per-seed dominance is reported at Table A3.)

A.4.4 TSTR Cross-Model Utility

Reviewer R1-M2 requested a Train-on-Synthetic, Test-on-Real (TSTR) evaluation as the downstream-utility complement to the distributional metrics. Figure A4 reports TSTR R^2 for a 1-step-ahead OD soft sensor (1-layer LSTM, hidden$= 32$) trained on synthetic windows from each of the 9 matched-budget generators (Pipeline B, 2000 epochs, 3 init seeds $\{40, 41, 42\}$).

A.5 Code and Data Availability

Complete code repository: `https://github.com/shawngibford/qgan`

Archived snapshot of the tagged release: deposited at Zenodo upon acceptance.

The complete code, trained models, and datasets have been released for transparency, independent verification, and acceleration of subsequent advances in quantum-enhanced chemical and biochemical engineering.

Artifact reference table. Every results JSON, code module, and documentation file cited elsewhere in this manuscript and supplement is enumerated in Table A9; in-line path strings throughout the paper have been replaced with the corresponding ID. Paths are relative to the project root in the released repository, all paths under the `results/`, `core/`, `docs/` trees.

Table A6: Per-seed OD-EMD values for all nine matched-budget generators ($n = 5$ seeds, Pipeline B, 2000 epochs). The four quantum entrants sit within a narrow 0.022-0.038 band on every seed (per-model means $\approx$ 0.028-0.031). The WGAN-CNN seed-42 cell (3.4338) is the headline outlier flagged in main §4.1 and Fig. A13f; the other four WGAN-CNN seeds sit in the 0.080-0.192 band. Even excluding WGAN-CNN, the surviving WGAN-MLP and WGAN-LSTM cluster means (0.0769 and 0.1177) sit above the quantum cluster on the 5-seed sample mean. The Mean column is the 5-seed sample mean.

Model	**Seed 42**	**Seed 43**	**Seed 44**	**Seed 45**	**Seed 46**	**Mean**
IQP:SEL (55p, Q)	0.0243	0.0339	0.0315	0.0259	0.0252	0.0282
V1 (75p, Q)	0.0229	0.0347	0.0331	0.0266	0.0244	0.0283
V2 (135p, Q)	0.0238	0.0339	0.0325	0.0266	0.0227	0.0279
V3 (75p, Q)	0.0262	0.0384	0.0344	0.0289	0.0262	0.0308
WGAN-MLP (74p, C)	0.0566	0.0756	0.1109	0.1006	0.0410	0.0769
WGAN-CNN (73p, C)	3.4338	0.0797	0.1661	0.1921	0.1227	0.7989
WGAN-LSTM (78p, C)	0.0292	0.1932	0.1095	0.0730	0.1834	0.1177
VAE (562p)	0.0207	0.0338	0.0327	0.0179	0.0237	0.0257
AR(2) (3p)	0.0252	0.0354	0.0321	0.0281	0.0247	0.0291

Table A7: Per-seed OD-DTW (`dtw_mean`) values for all nine matched-budget generators ($n = 5$ seeds, Pipeline B, 2000 epochs). The four quantum entrants sit inside a tight 0.32-0.45 per-seed band on every seed; the WGAN-MLP and WGAN-LSTM cluster means (0.91 and 0.60) sit substantially above the quantum cluster, and the WGAN-CNN seed-42 cell (28.4041) is the headline outlier consistent with the same model's OD-EMD outlier and with the critic-loss drift in Fig. A13f. The two non-adversarial baselines (VAE 0.31, AR(2) 0.37) sit inside the quantum range. The Mean column is the 5-seed sample mean.

Model	**Seed 42**	**Seed 43**	**Seed 44**	**Seed 45**	**Seed 46**	**Mean**
IQP:SEL (55p, Q)	0.3562	0.3368	0.3761	0.3926	0.3894	0.3702
V1 (75p, Q)	0.3479	0.3370	0.3341	0.4045	0.3222	0.3491
V2 (135p, Q)	0.3230	0.3316	0.3563	0.3494	0.3042	0.3329
V3 (75p, Q)	0.3954	0.4025	0.4239	0.4541	0.3723	0.4096
WGAN-MLP (74p, C)	1.0748	1.2247	0.7662	0.7674	0.7410	0.9148
WGAN-CNN (73p, C)	28.4041	1.4428	1.6579	1.7460	1.7030	6.9907
WGAN-LSTM (78p, C)	0.4660	0.8164	0.4787	0.5706	0.6520	0.5967
VAE (562p)	0.2836	0.3038	0.2941	0.3719	0.2812	0.3069
AR(2) (3p)	0.3717	0.3510	0.3990	0.4113	0.3204	0.3707

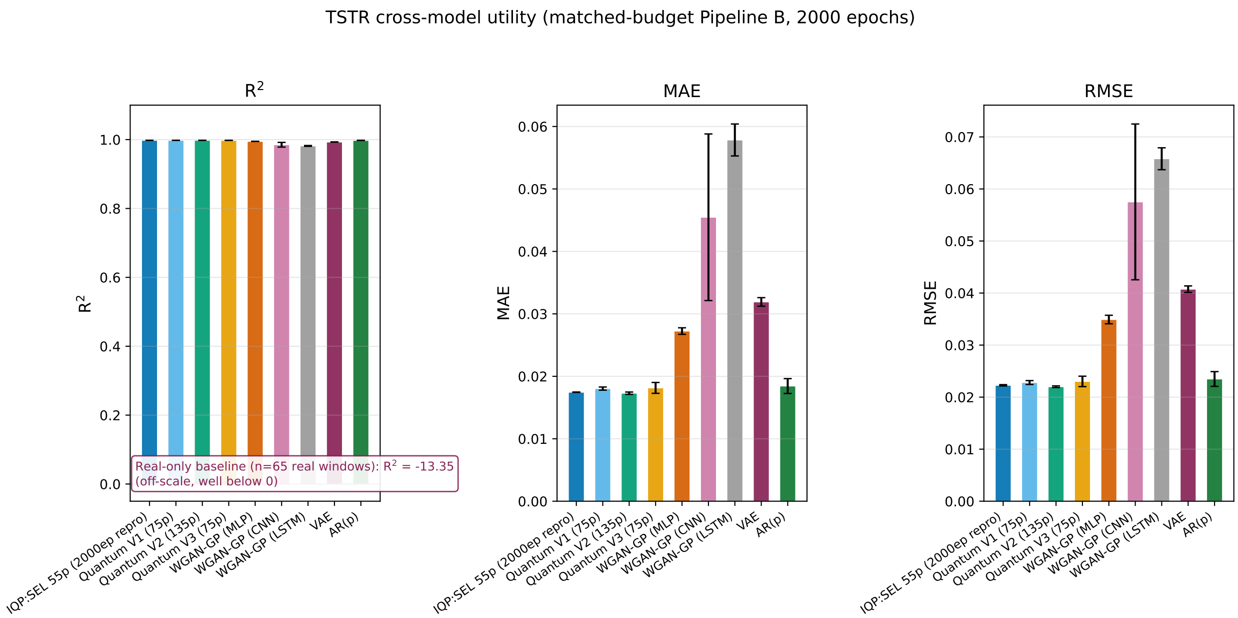


Figure A4: TSTR cross-model utility evaluation: 1-step-ahead OD soft sensor (1-layer LSTM, hidden= 32) trained on 19200 synthetic windows from each generator (Pipeline B, 2000 epochs, 3 init seeds $\{40, 41, 42\}$), evaluated on $n = 320$ held-out real windows. All 9 matched-budget generators converge to $R^2 \approx 0.99$, dramatically above the real-only baseline of $R^2 = -13.35$ ($n = 65$ real training windows; the small real set fails to support soft-sensor training). The cross-generator convergence is *structural* (caveat): the cumulative-sum back-transform from log-returns to OD mathematically encodes near-perfect lag-1 autocorrelation, so a soft sensor trained on Pipeline-B-derived synthetic OD essentially learns the persistence forecast, which is near-optimal on real OD regardless of which generator produced the underlying log-returns. The Pipeline-B TSTR therefore does *not* discriminate among generators (R1-M2 disclosure); meaningful TSTR discrimination would require multivariate process data where soft-sensor utility depends on cross-channel conditional structure.

Table A8: Per-seed log-return EMD (LR-EMD) values for all nine matched-budget generators ($n = 5$ seeds, Pipeline B, 2000 epochs). The four quantum entrants sit in the 0.0030-0.0077 per-seed band on every seed (per-model means 0.0040-0.0050); the three WGAN-kind classical adversarial baselines sit substantially above (WGAN-MLP mean 0.0444, WGAN-CNN mean 0.1286, WGAN-LSTM mean 0.0244). AR(2) is the only classical baseline that beats the quantum cluster on this metric (closed-form Yule-Walker fit on the marginal log-return distribution is structurally near-optimal). The Mean column is the 5-seed sample mean.

Model	**Seed 42**	**Seed 43**	**Seed 44**	**Seed 45**	**Seed 46**	**Mean**
IQP:SEL (55p, Q)	0.0034	0.0054	0.0043	0.0041	0.0030	0.0040
V1 (75p, Q)	0.0046	0.0044	0.0033	0.0041	0.0042	0.0041
V2 (135p, Q)	0.0044	0.0045	0.0044	0.0043	0.0044	0.0044
V3 (75p, Q)	0.0051	0.0030	0.0077	0.0042	0.0050	0.0050
WGAN-MLP (74p, C)	0.0354	0.0560	0.0444	0.0444	0.0420	0.0444
WGAN-CNN (73p, C)	0.2365	0.1062	0.0755	0.1218	0.1029	0.1286
WGAN-LSTM (78p, C)	0.0238	0.0259	0.0269	0.0153	0.0301	0.0244
VAE (562p)	0.0158	0.0158	0.0159	0.0158	0.0159	0.0158
AR(2) (3p)	0.0028	0.0030	0.0030	0.0031	0.0029	0.0029

The number-provenance gate (C1) audits every literal in the manuscript, supplementary, and rebuttal against the results corpus at submission time; the gate schema, the corpus snapshot, and the per-literal resolution manifest are part of the tagged release.

Quantum Circuit Specifications:

The quantum circuit employs a dual-stage encoding strategy as described in Section 3 of the main text:

- 5 qubits arranged in three layers
- IQP (Instantaneous Quantum Polynomial) encoding
- Strongly entangling layers with $\mathrm{Rot}(\phi, \theta, \omega)$ gates
- Range-based CNOT gates: $r = (\text{layer mod } (n_{\text{qubits}} - 1)) + 1$
- Enhanced Pauli measurements (X and Z expectation values)
- Total parameters: 55 (5 for IQP encoding, 45 for the three strongly entangling layers, 5 for measurement-preparation rotations)

A.6 Additional Figures

Quantum Circuit Family: Headline Generator and Ablation Variants

The headline quantum generator (`iqp_sel_55`) was selected from a four-member ansatz family that varies layer depth, parameter count, and entangler topology while holding the qubit count (5) and the IQP-encoding plus strongly-entangling-layer (SEL) backbone fixed. The

Table A9: Artifact reference table. **R**-rows are results files (JSON / CSV) under `results/`; **C**-rows are code modules + executable scripts under `core/` and `run_*`; **D**-rows are documentation files. All paths relative to the project root in the released repository.

ID	Path	Purpose / contents
Results aggregates (main paper figures + tables)		
R1	`matched2000_dualscale.json`	Main Table 2 + Figs 3-5 aggregates; dual-scale per-model + per-seed values
R2	`distribution_emd.json`	Per-model distributional EMD on log-return + OD scales
R3	`welch_pairwise.json`	Pairwise Welch t-tests on EMD (§A.4)
R4	`cross_model_emd.json`	Fig 5 EMD overlay across all 9 models
R5	`cross_model_dtw_dualscale.json`	Fig 3 DTW dual-scale
R6	`cross_model_acf_overlay.json`	Fig 4 lag-1 ACF overlay
R7	`model_info.json`	Master hyperparameter + metadata record
R8	`tstr_matched2000.json`	TSTR cross-model utility (§A.4)
R9	`shot_noise_sensitivity.json`	Shot-noise sensitivity sweep (§A.14)
R10	`noise_model_sensitivity.json`	Depolarising + amplitude-damping noise sweep (§A.14)
R11	`transform_ablation/metrics.csv`	Preprocessing-pipeline ablation metrics
R12	`transform_ablation/tstr_lite.json`	Preprocessing-pipeline TSTR-lite
R13	`figures/training_convergence_all_models.json`	In-loop OD-EMD convergence across all 7 WGAN-GP generators
Per-model + per-seed JSONs		
R14	`matched2000/runs/<model>/<seed>/metrics.json`	Per-seed training-loop trajectories
R15	`figures/loss_<model>.json`	Per-model loss-figure data + fit summaries
R16	`figures/reconstruction_<model>.json`	Per-model reconstruction-figure data
R17	`figures/training_progression.json`	Quantum circuit training-progression (§A.8)
R18	`figures/entanglement_trajectory.json`	Quantum entanglement entropy over training (§A.8)
R19	`figures/param_trajectory.json`	PQC parameter trajectories over training (§A.8)
R20	`figures/circuits/<name>.json`	Locked circuit configurations (per ansatz)
Code modules + executable scripts		
C1	`verify_number_provenance.py`	Number-provenance audit gate (every literal → JSON)
C2	`core/eval.py`	Canonical metric implementations (DTW, EMD, JSD, moments)
C3	`core/data.py`	Data loading, log-return + Lambert W transforms
C4	`run_baselines.py`	Classical baseline orchestrator
C5	`run_welch_aggregator.py`	Welch-table aggregator script
Documentation files + architecture records		
D1	`docs/methods_full.md`	Full methods documentation
D2	`docs/circuit_atlas.md`	Per-ansatz circuit gate sequences
D3	`results/classical_architectures.json`	Critic + classical-generator per-layer architecture record
D4	`README.md`	Script entry-points + reproduction instructions

three additional variants (V1, V2, V3) are the ablation comparators reported in the main-text expressibility-versus-trainability analysis (§3.2 of the main text). Figure A5 shows the rendered gate sequence for each of the four circuits; per-circuit specifications follow below the figure.

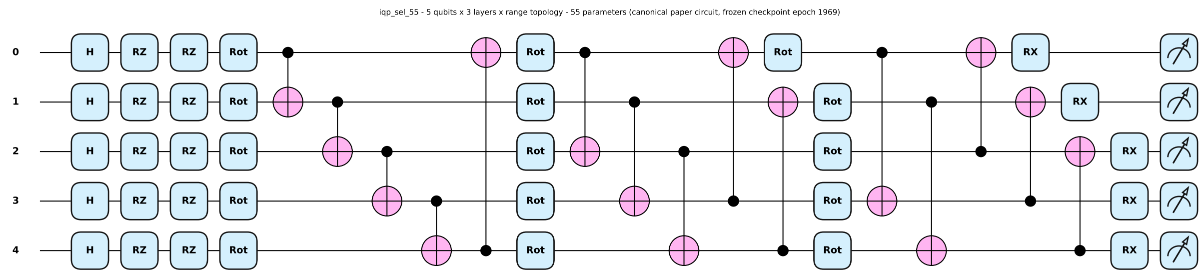
Ansatz variants. The four ansatz variants compared at matched-budget are: **IQP:SEL** (55 parameters, depth 3, range entangler — the reported headline generator), **V1** (75p, depth 4, range entangler — depth-increment control), **V2** (135p, depth 8, range entangler — deepest range-entangled variant), and **V3** (75p, depth 4, linear entangler — isolates entangler topology from depth at fixed parameter count). Each variant is depicted as a panel of Figure A5, with per-variant gate-sequence + qubit topology + encoding/measurement basis recorded in D2. Under the matched-budget contract, the smaller 55-parameter IQP:SEL circuit was not outperformed in fidelity by any of V1-V3, motivating its selection as the reported configuration (see the headline LR-DTW dominance reported in main-text §4.1) [56, 57].

This figure illustrates the decision-tree triage workflow outlined as future work in the Outlook subsection of Section 4 in the main text, sketching how sensor-feasibility, mechanistic, data-driven, and quantum-synthetic options could be organised under varying data-availability constraints. It is presented as an organising concept and is not evaluated empirically in this paper.

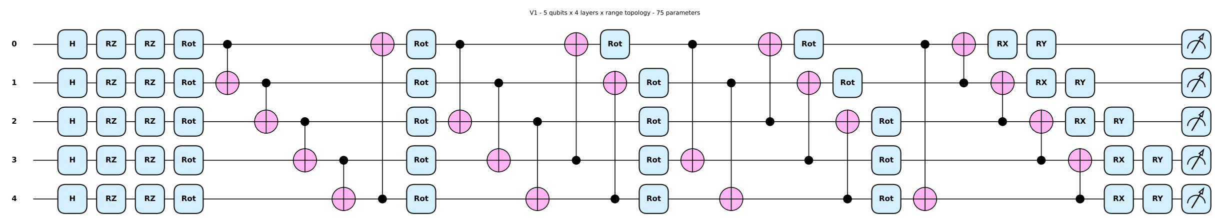
This schematic depicts the photobioreactor experimental setup described in Section 3.3, showing the sensor and actuator configuration used for data acquisition.

A.7 Data Transformation Details

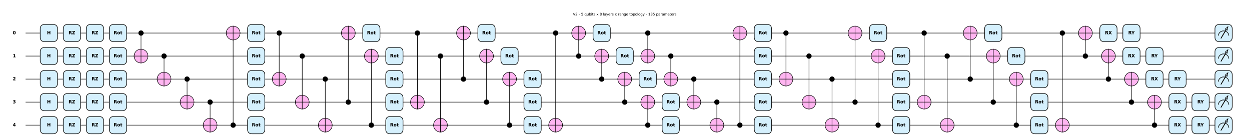
The transformation from optical density measurements to log differences is motivated by the bioprocess itself: for an exponentially growing culture the optical density follows $\mathrm{OD}_t \approx \mathrm{OD}_0\, e^{\mu t}$, so the log difference between successive samples is a direct discrete estimate of the instantaneous specific growth rate. For optical density measurements OD_t at time t:



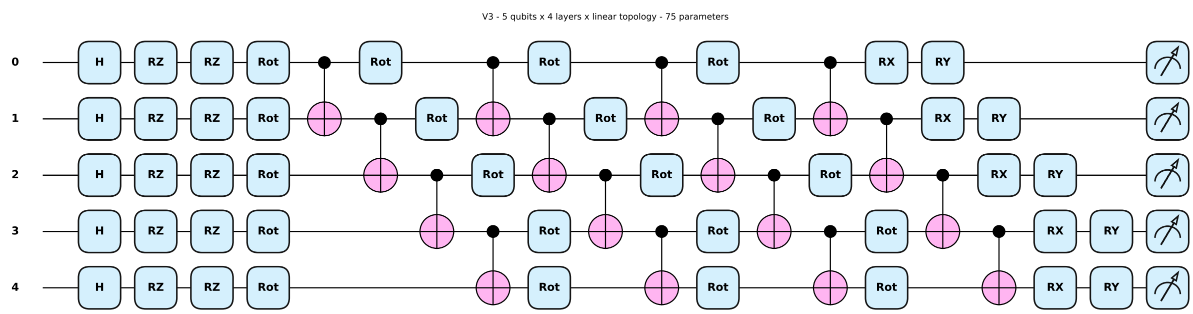
(a) IQP:SEL: 55 parameters, depth 3, range entangler (headline circuit).



(b) V1: 75 parameters, depth 4, range entangler.



(c) V2: 135 parameters, depth 8, range entangler.



(d) V3: 75 parameters, depth 4, linear entangler.

Figure A5: Rendered gate sequences for the four 5-qubit PQC generators evaluated in this study. All variants share an IQP encoding stage followed by strongly-entangling layers (SEL) with Rot(ϕ, θ, ω) rotations and CNOT entanglers; they differ in layer depth, total trainable-parameter count, and entangler connectivity (range versus linear). The headline circuit (a, IQP:SEL, 55 parameters, depth 3, range topology) is the smallest in the family. Variants V1-V3 span the expressibility-versus-trainability trade-off explored in the main-text §3.2 ablation. Rendered with PennyLane's `qml.draw_mpl` from the locked circuit configurations.

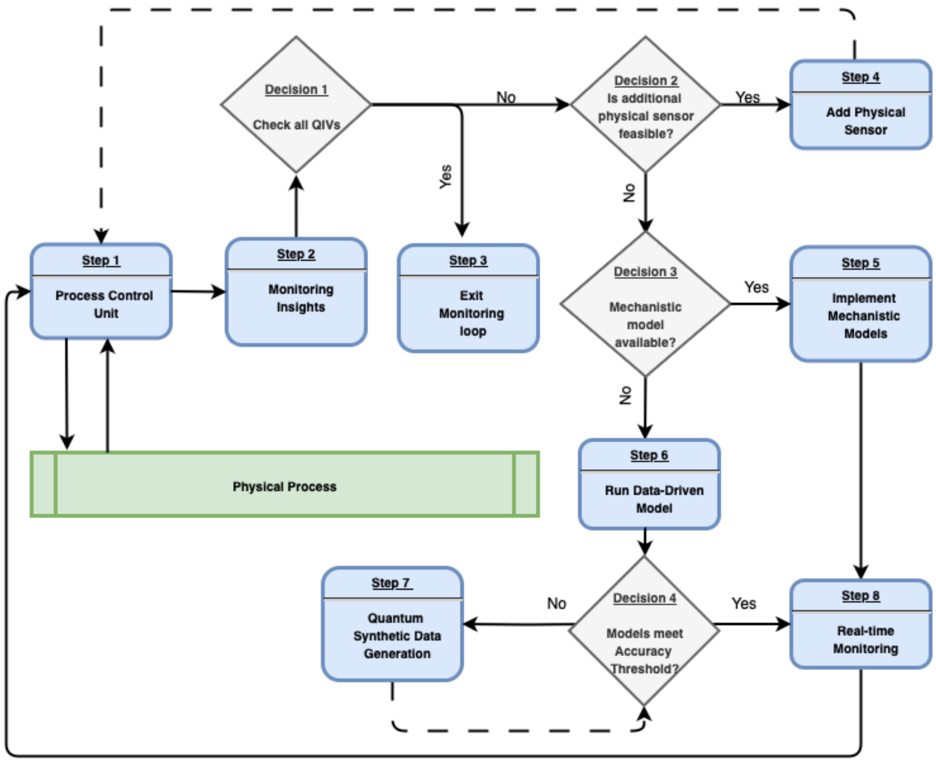


Figure A6: Decision-tree triage schematic for bioprocess monitoring, outlined as future work in the Outlook subsection of Section 4 in the main text. Not an empirical contribution of the present study.

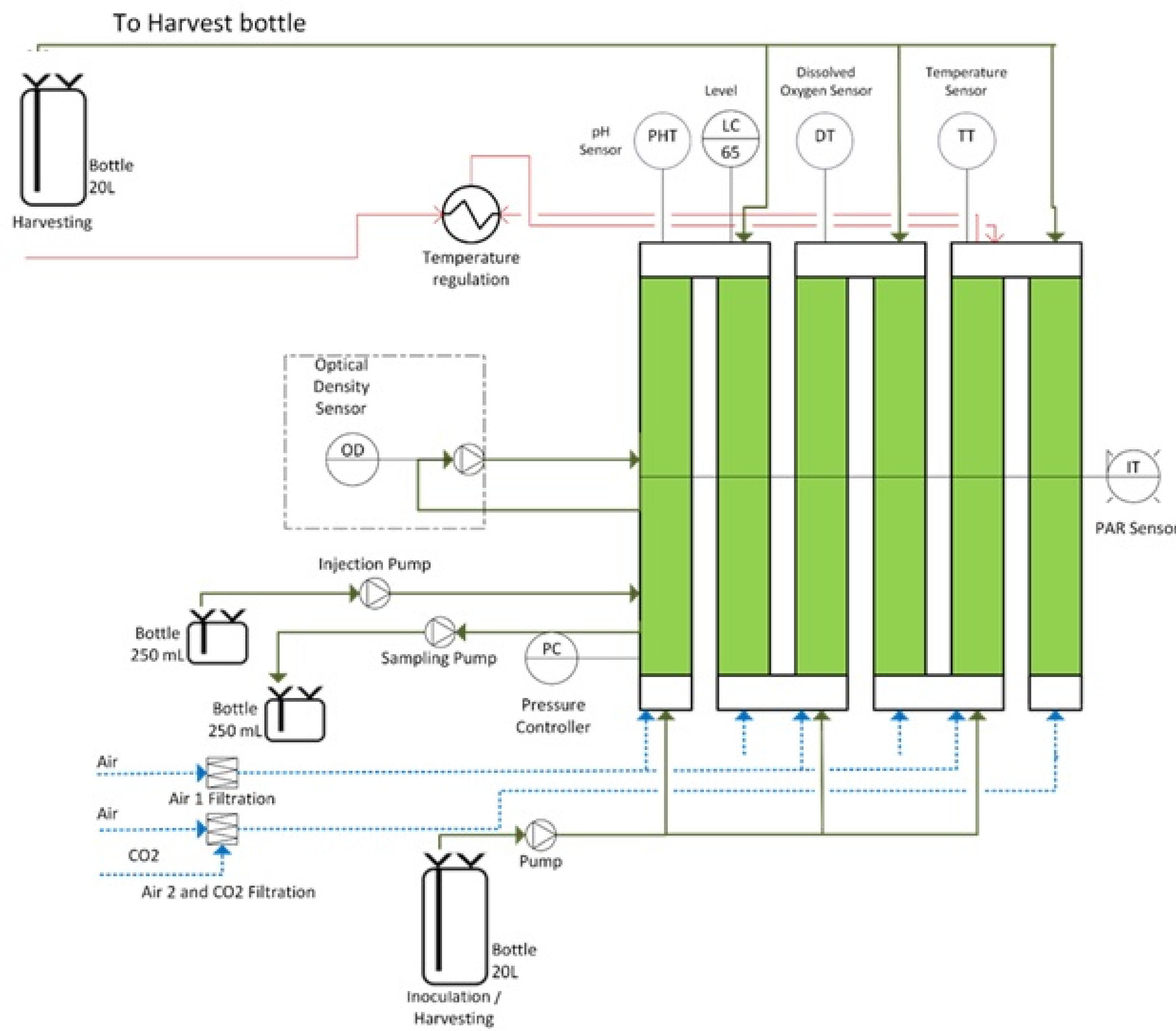


Figure A7: Schematic of the 20 L LUCY® photobioreactor used in this study, with its sensors and actuators.

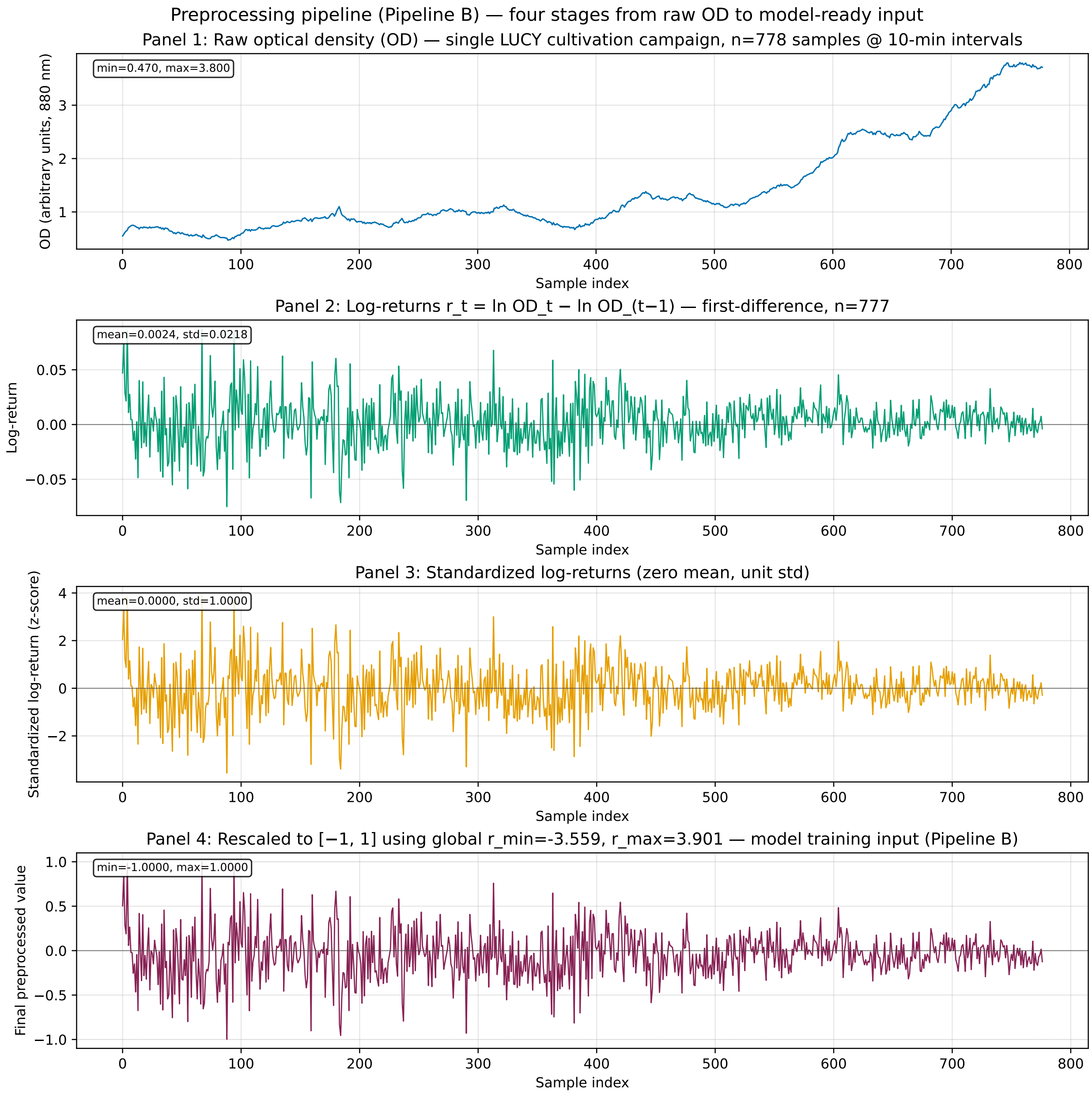


Figure A8: Four-stage preprocessing pipeline (Pipeline B, native): raw OD ($n = 778$), log-returns $r_t = \ln \mathrm{OD}_t - \ln \mathrm{OD}_{t-1}$ ($n = 777$), standardized ($\mu = 0$, $\sigma = 1$), and linearly rescaled to $[-1, 1]$ using the global min and max of the standardized log-return series. The trained generators operate on the final-stage rescaled log-return windows; all matched-budget evaluation results in Section 4 of the main text are reported under this pipeline.

$$r_t = \ln(\mathrm{OD}_t) - \ln(\mathrm{OD}_{t-1}) \;\approx\; \mu_t \, \Delta t$$

where μ_t is the local specific growth rate over the sampling interval Δt. Working in r_t rather than raw OD has a direct bioprocess interpretation and three practical benefits for this growth-rate signal:

- **Growth-rate interpretability:** each r_t is (up to the sampling interval) the specific growth rate, the quantity of physiological interest in cultivation monitoring, rather than an arbitrary attenuation reading
- **Scale/stage invariance:** the log difference is invariant to the absolute OD level, so early-exponential and high-density phases are placed on a comparable scale, which stabilises learning across the growth curve
- **Additivity:** growth rates over consecutive intervals sum to the net log-growth over the total span, matching how cumulative biomass increase is reported

The finance literature uses the same transform for unrelated reasons; here the justification is physiological (growth-rate estimation), not financial.

Normalization:

$$r_t^{\mathrm{NORM}} = \frac{r_t - \mu}{\sigma} \tag{A4}$$

This brings all values to a consistent scale with mean zero and standard deviation one, which is essential for quantum circuit encoding.

Rescale to $[-1, 1]$: The standardized log-returns are linearly rescaled to the closed interval $[-1, 1]$ using the global minimum and maximum of the standardized series,

$$r_t^{[-1,1]} = -1 + 2\,\frac{r_t^{\mathrm{NORM}} - r_{\min}^{\mathrm{NORM}}}{r_{\max}^{\mathrm{NORM}} - r_{\min}^{\mathrm{NORM}}}, \tag{A5}$$

to match the bounded output range of the quantum circuit observables.

Preprocessing ablation: why no Lambert W transform. An earlier version of this pipeline (Pipeline C in our terminology) applied an inverse Lambert W heavy-tail correction [62] between the standardization and rescaling steps. A 5-seed ablation at 1000 epochs (Figure A9) compared three preprocessing chains: *Pipeline A* (raw min-max-normalized OD; control), *Pipeline B* (log-returns standardized + rescaled, as above), and *Pipeline C* (log-return → standardize → inverse Lambert W → rescale to $[-1, 1]$; i.e., Pipeline B with an inverse Lambert W step inserted between standardization and rescaling). The three pipelines were trained under identical 5-qubit, 3-layer, 55-parameter IQP:SEL generators. Pipeline A trains successfully but produces clearly worse OD-scale fidelity (EMD 1.0516 ± 0.0007 vs 0.0276 for B and 0.0261 for C; OD-ACF lag-1 -0.094 vs real-data reference 0.456). Pipelines B and C are within sampling noise of each other on every OD-scale metric (EMD 0.0276 vs 0.0261; OD-ACF lag-1 0.696 vs 0.697; OD-DTW mean 0.30 vs 0.41; TSTR-lite R^2 0.994 vs 0.994). Per decision D-10-05 we therefore select Pipeline B for the matched-budget runs reported in the main text, dropping the Lambert W step on two grounds: (i) it does not improve OD-scale fidelity over Pipeline B; (ii) it directly addresses reviewer concern R1-M3, which flagged that the Lambert W step over-Gaussianizes the input distribution and could trivialise the generative task. Full per-seed metrics and the ablation summary are released with the artifact set (R11 and R12).

Sample-space convention and inference-time correction. All WGAN-trained generators in this study (the four quantum variants IQP:SEL, V1, V2, V3 and the three classical WGAN-GP baselines wgan_mlp, wgan_cnn, wgan_lstm) export samples with their $[-1, +1]$ generator outputs multiplied by 0.1 at the save site; the on-disk `samples.npy` bundles are therefore stored in $[-0.1, +0.1]$ space. This convention is inherited from an earlier reference implementation (the relevant code comment magnitude-matches the $[-1, +1]$ generator output to the empirical range of unstandardized log-returns, $[-0.08, 0.09]$). The convention is retained at the training/sample-export sites so that the released `samples.npy` bundles remain byte-stable across runs and consistent with every saved checkpoint; modifying the training-side scaling would invalidate every released artifact and require re-running every 2000-epoch run. The downstream inverse formula $r_{\text{norm}} = ((s+1)/2)(r_{\max} - r_{\min}) + r_{\min}$ assumes $s \in [-1, +1]$, so every evaluation script applies a $\times 10$ correction at the `samples.npy` load boundary before invoking the inverse formula. The correction is inference-only,

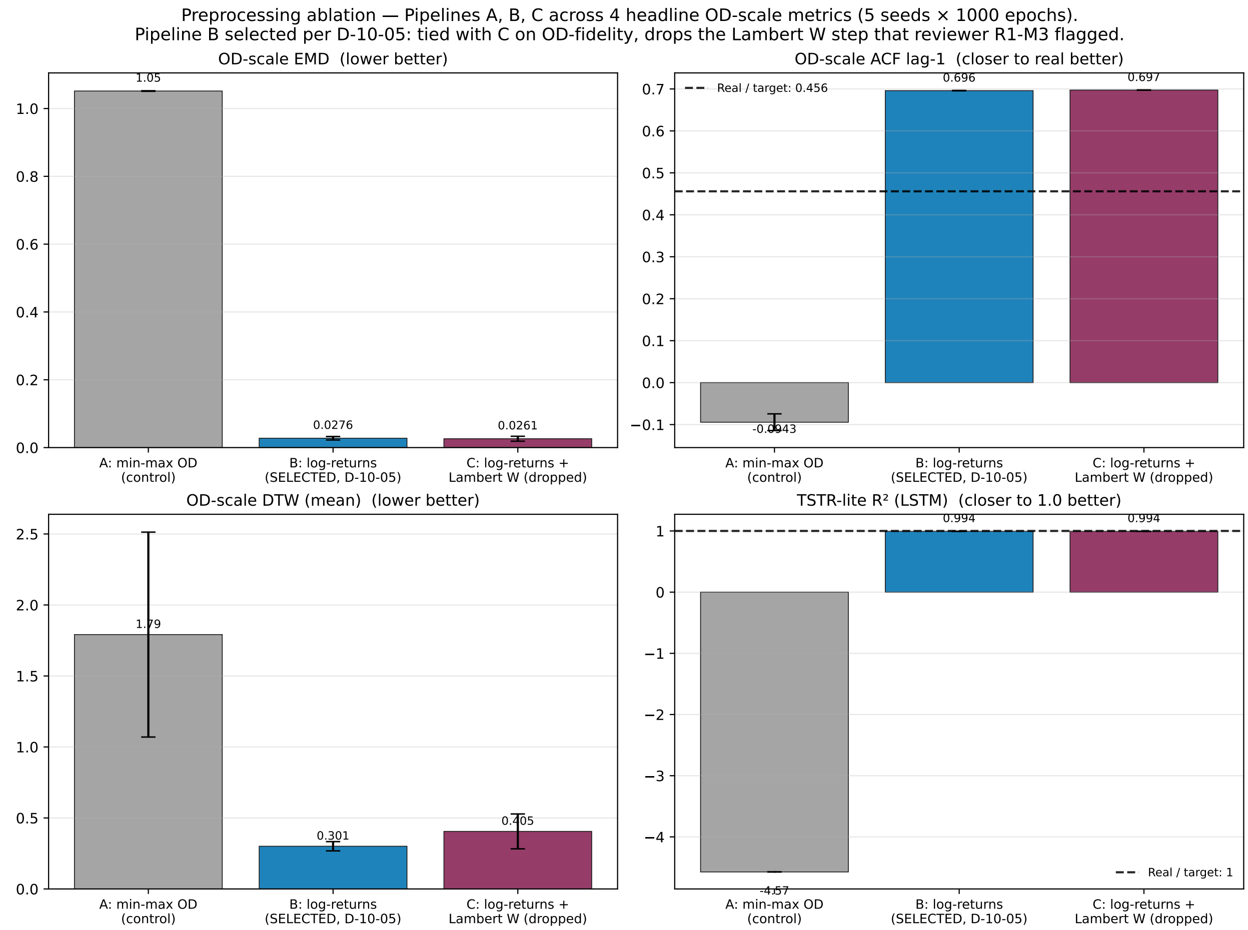


Figure A9: Preprocessing ablation across three pipelines (A: raw min-max OD; B: log-returns standardized + rescaled to $[-1, 1]$, selected; C: log-return $\rightarrow$ standardize $\rightarrow$ inverse Lambert W $\rightarrow$ rescale to $[-1, 1]$, dropped). Bars are 5-seed mean $\pm$ sample standard deviation under identical 5-qubit, 3-layer, 55-parameter IQP:SEL generators trained at 1000 epochs. Pipelines B and C are within sampling noise of each other on every OD-scale metric; Pipeline B was selected per decision D-10-05.

applied symmetrically across the quantum and classical WGAN-trained set, and gated by `model_kind` membership in the WGAN-trained set. VAE and AR(2) are deliberately excluded: they are trained without the $\times 0.1$ sample-export step and their on-disk samples are already in the canonical $[-1, +1]$ space. Every matched-budget headline metric reported in §4 of the main text is computed under this $\times 10$-corrected inverse pipeline.

A.8 Model Architecture Details and Quantum Circuit Introspection

A.8.1 Critic network architecture

The critic model is a convolutional neural network that processes sequential data to distinguish between real and generated samples, following the WGAN-GP architecture [34] with adaptations for time-series.

The critic network has a total of 250881 trainable parameters. This same critic was used for all WGAN-GP generators (quantum and classical) under the matched-budget protocol defined in main-text Section 3.1; only the generator varies between quantum and classical entrants, isolating the effect of the quantum circuit on the generated distribution.

The critic is a standard WGAN-GP 1D-CNN: three Conv1D blocks with LeakyReLU activations followed by a flatten-and-dense head with dropout and an unbounded scalar output, consistent with the gradient-penalty WGAN formulation [34, 35]. The full per-layer specification (filter counts, kernel sizes, activation slope, dropout rate, and dense width) is recorded in the released artifact corpus at D1 and D3; both are provenance-verified and tracked in the matched-budget reproducibility bundle.

A.8.2 Quantum circuit introspection diagnostics

This section addresses reviewer comment R2-6 (“what is the quantum circuit actually learning?”) with three introspection diagnostics produced under Phase 13 of the revision. The three diagnostics are reported on a single representative trajectory (seed 42, Pipeline B, snapshot epochs $\{0, 250, 500, 750, 1000\}$); the matched-budget quantitative evaluation in main §4.1 is over the full 5-seed 2000-epoch protocol.

A.9 Per-Model Training Loss Diagnostics

Complementing the in-loop OD-EMD convergence figure in main §3.1 (Figure 2), Figure A13 reports the raw per-model training losses (single representative seed 42, 2000 epochs, Pipeline B). The panels surface three qualitatively distinct training regimes.

WGAN-GP critic stability across the cohort. For all 7 WGAN-GP generators (4 quantum + 3 classical adversarial), the critic loss descends from $\sim$+2 at initialisation to a stable small-magnitude negative band within $\sim$100 eval steps; the generator loss oscillates within a bounded range, with no NaN, no divergence, and no critic saturation across any of the 35 runs (7 models $\times$ 5 seeds). The IQP:SEL panel (Figure A13a) is representative: critic loss settles near -0.65, generator loss oscillates between 0.1 and 1.2.

WGAN-CNN training instability (quantum-classical contrast). The WGAN-CNN panel (Figure A13f) shows a qualitatively different signature: the critic loss bottoms near -3.0 by eval step ~ 50 and then drifts back toward -1.2 over the remainder of training; the critic progressively loses discriminative power against the WGAN-CNN generator. The 4 quantum and 2 other classical adversarial generators do not exhibit this drift. The signature is consistent with the seed-42 OD-EMD outlier reported in main §4.1 and with the dip in Figure 2: WGAN-CNN is intermittently fragile at the matched-budget contract while the quantum cluster is stable across all 5 seeds.

VAE convergence pattern (regularization collapse). The VAE panel (Figure A13h) shows ELBO and reconstruction terms collapsing to ~ 0.058 within the first $\sim$50 eval steps and remaining flat thereafter, while the KL divergence term collapses essentially to zero in the same window and never recovers. This regularization-term collapse is the training-side signature of the degenerate generation regime characterized in main §4.1: the encoder finds a near-zero-KL solution (posterior matched to the prior) and the decoder produces real-matching one-step marginals while losing the temporal structure exposed by the lag-1 ACF -0.648 at evalua-

Figure A10: Generated log-return distribution snapshots across training: 4 generators (quantum IQP:SEL 55p; wgan_mlp 74p; wgan_cnn 73p; wgan_lstm 78p) at snapshot epochs $\{0, 250, 500, 750, 1000\}$. Each cell overlays the synthetic distribution against the real reference. The quantum row qualitatively tracks the real reference more closely than the classical rows by epoch 1000, anticipating the LR-DTW + lag-1 ACF dominance observed at the matched-budget 2000-epoch evaluation reported in main §4.1. Single seed 42, Pipeline B.

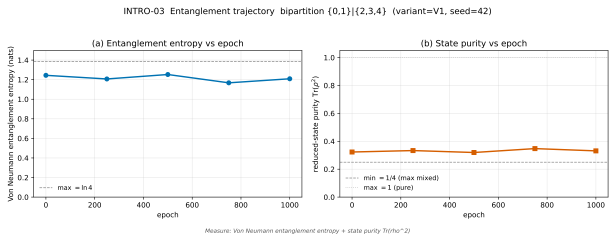


Figure A11: Entanglement entropy trajectory of the quantum generator during training (variant V1, 4 layers, range topology, seed 42). Bipartition $\{0,1\} \mid \{2,3,4\}$. Top: Von Neumann entanglement entropy across snapshot epochs $\{0, 250, 500, 750, 1000\}$, values in $[1.17, 1.25]$ against the reference bound $\ln 4 \approx 1.386$ for maximal entanglement on this bipartition. Bottom: state purity $\mathrm{Tr}(\rho^2) \in [0.32, 0.35]$ against the reference bound 0.25 for the maximally mixed state. The trajectory shows the generator maintains a highly entangled state throughout training without collapsing to a product state, providing the quantum-resource signal R2-6 requested.

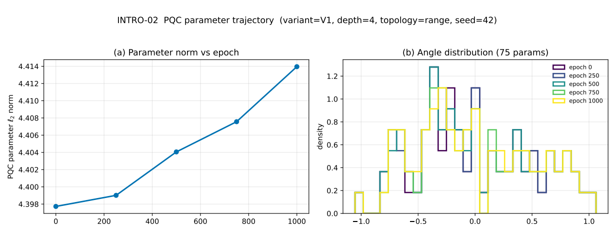


Figure A12: PQC parameter trajectory during training (variant V1, 4 layers, range topology, 75 parameters, seed 42). Top: L_2-norm of the parameter vector across snapshot epochs $\{0, 250, 500, 750, 1000\}$; values stably bounded between ≈ 4.40 and 4.41 (no divergence, no collapse to zero). Bottom: distribution of all 75 individual rotation angles at each snapshot epoch, with the angle diversity maintained throughout training rather than collapsing into a narrow band. Together the panels show the optimizer explores the parameter space without pathological behavior.

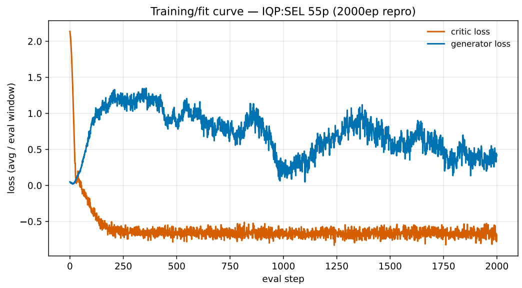
tion time.

AR(2): closed-form fit. AR(2) is not included in the loss grid because it is fit in closed form via `numpy.linalg.lstsq` on the training log-return windows (residual variance $\hat{\sigma}^2 = 0.057$, sample mean -0.003, sample standard deviation 0.242) and therefore has no epoch-wise loss trajectory. The fit-summary statistics are recorded in the `fit_summary` block of R15 for AR(2).

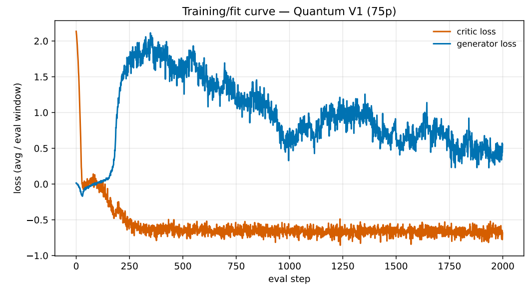
No training pathology detected at the matched-budget contract. Across all 9 generators (or 35 individual WGAN-GP runs), no run diverged, NaN-ed, or required intervention; the WGAN-CNN critic-drift is the only training-side anomaly and is consistent with that model's known fragility at the matched-budget parameter count of 73.

A.10 Per-Model Reconstruction Overlays

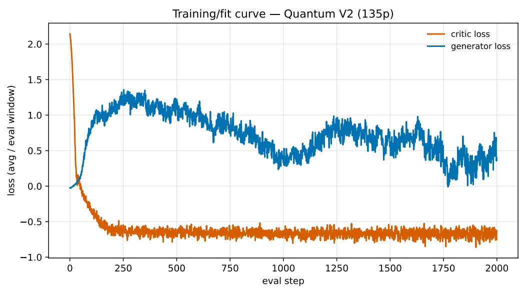
Figure A14 provides a per-model time-series view of generation fidelity across all nine matched-budget generators (seed 42), complementing the per-model loss diagnostics of §A.9 and the cross-model distributional metrics of main-text §4. Each subpanel con-



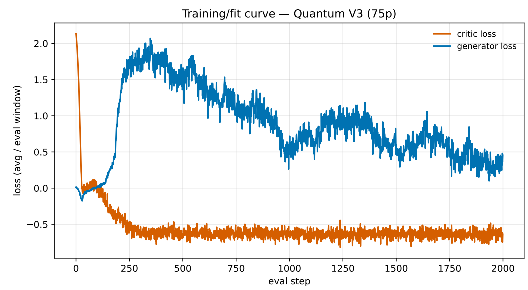
(a) IQP:SEL (55p, Q)



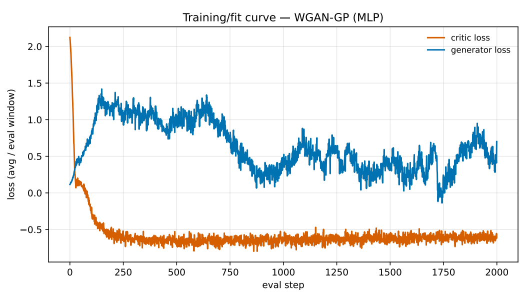
(b) V1 (75p, Q)



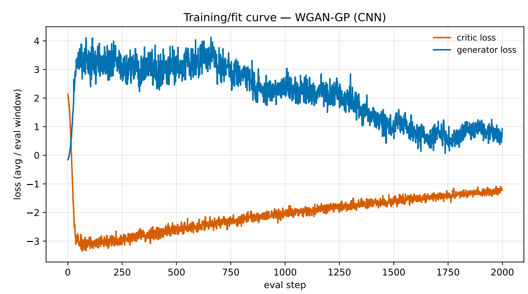
(c) V2 (135p, Q)



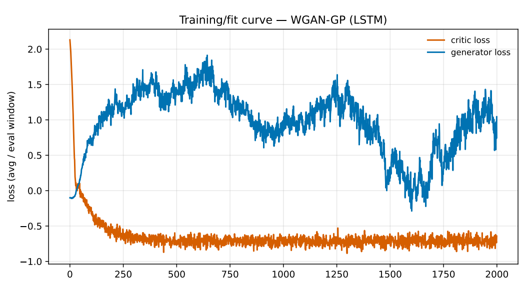
(d) V3 (75p, Q)



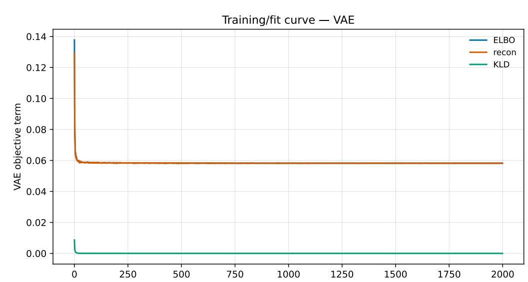
(e) WGAN-MLP (74p, C)



(f) WGAN-CNN (73p, C)



(g) WGAN-LSTM (78p, C)



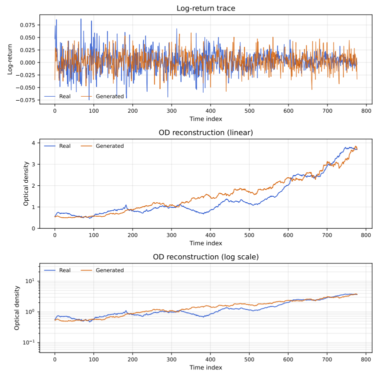
(h) VAE (562p, ELBO+recon+KLD)

Figure A13: Per-model training loss trajectories at the matched-budget contract (single representative seed 42, 2000 epochs, Pipeline B). Panels (a)-(d) show the 4 quantum WGAN-GP variants; panels (e)-(g) show the 3 classical adversarial WGAN-GP baselines; panel (h) shows the VAE (single-Adam ELBO loop; the only non-adversarial model with a per-epoch trajectory). Q = quantum, C = classical. AR(2) is fit in closed form via `lstsq` and has no epoch-wise loss trajectory ($\hat{\sigma}^2 = 0.057$ reported in text).

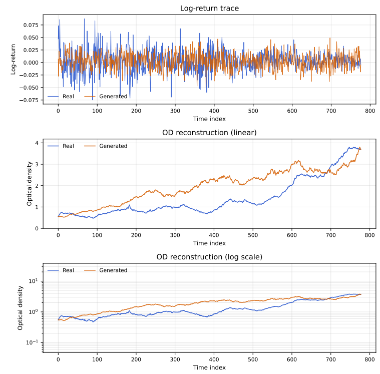
sists of three vertically stacked sub-plots (log-delta comparison, OD reconstruction on linear scale, OD reconstruction on log scale) per the protocol detailed in the umbrella caption. The visualisation protocol differs from the canonical matched-budget per-window evaluation of main-text §4.1: synthetic samples are reconstructed as a 777-point contiguous sequence anchored at $OD_{real}[0]$, rather than aggregated over the per-window random anchors that the fidelity metrics use.

Mean-shift visualization protocol. A residual generator mean-bias in standardized log-return space persists across the matched-budget cluster (raw per-step gen mean ranges from -0.010 to 0.137 vs. real Pipeline-B mean ≈ 0.0025). Under the long-horizon overlay protocol used here – 777 contiguous integration steps anchored at $OD_{real}[0]$ – this per-step bias compounds exponentially: V1/V2/V3 overshoot real OD by $\sim$ 8-12×; WGAN-CNN reaches $\sim 10^{46}$. To prevent the compound-bias artifact from swamping the structural-fidelity signal (variance, autocorrelation, oscillation shape) that this atlas is intended to convey, the generated log-return sequence is *mean-shifted* prior to cumulative integration: the gen mean is subtracted and the real mean is added, leaving the per-step variance, autocorrelation, and higher moments *unchanged*. The exact shift is recorded per model in `drift_correction` of the companion JSON. This visualization choice is local to this atlas; the headline matched-budget fidelity metrics reported in main-text §4.2 (EMD, DTW, predictive/discriminative scores) are computed on un-corrected log-returns and are unaffected.

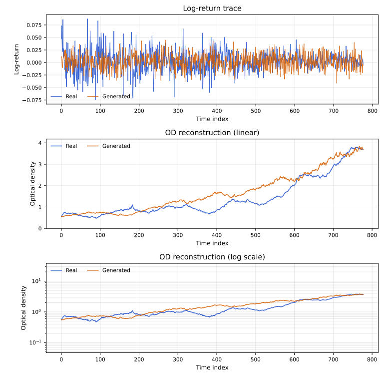
Un-shifted companion (reviewer-facing). For transparency on the visualization choice made in Figure A14, Figure A15 shows the *un-shifted* cumulative reconstruction for four representative models: two quantum (IQP:SEL, V1) and two extreme classical (WGAN-CNN, WGAN-LSTM). The artifact is not architecture-specific – IQP:SEL overshoots by $\sim$5× and V1 by $\sim$12× from per-step biases on the order of 0.002-0.003 – and the catastrophic classical cases fail in opposite directions (WGAN-CNN runs away to $\sim 10^{46}$; WGAN-LSTM collapses to $\sim 10^{-4}$), ruling out any "classical-models-



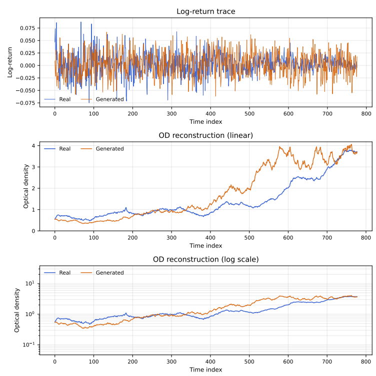
(a) IQP:SEL (55p, Q)



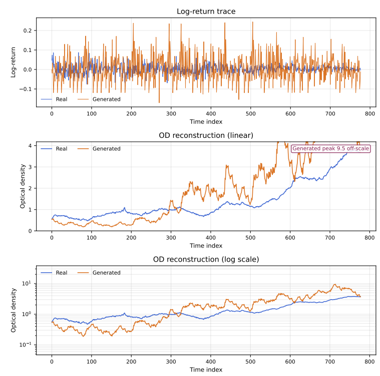
(b) V1 (75p, Q)



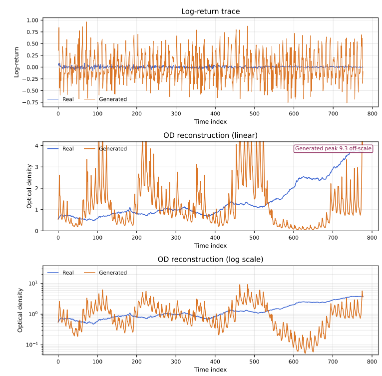
(c) V2 (135p, Q)



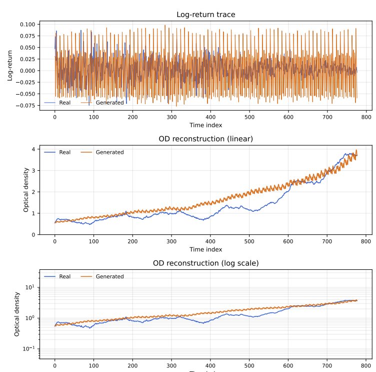
(d) V3 (75p, Q, linear)



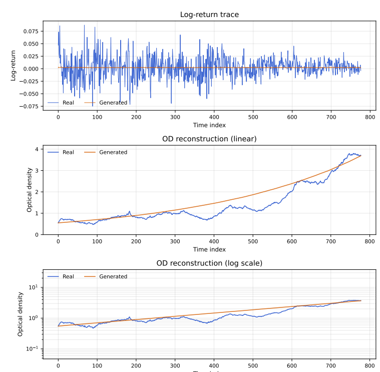
(e) WGAN-MLP (74p)



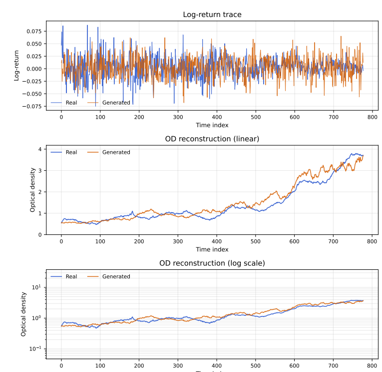
(f) WGAN-CNN (73p)



(g) WGAN-LSTM (78p)



(h) VAE (562p)



(i) AR(2) (3p)

Figure A14: Per-model reconstruction overlay atlas, seed 42, 2000ep matched budget. Each panel: top sub-plot = log-delta comparison (777 points, real blue vs. generated orange); middle = OD reconstruction on linear scale (778 points); bottom = OD reconstruction on log scale (778 points). Sample construction: contiguous 777-point log-delta sequence from each model's released sample bundle, un-rescaled then un-standardised via the saved $r_{\min}, r_{\max}, \mu, \sigma$; mean-shifted to the real log-return mean (see protocol paragraph above); OD reconstructed by anchoring at $\mathrm{OD}_{\mathrm{real}}[0]$ and integrating. Panel (f) is the WGAN-CNN seed-42 OD-EMD outlier (OD-EMD = 3.4338 vs. the other four seeds in 0.080-0.192), disclosed at face value in main-text §4.1. Panel (h) is the VAE degenerate-generation regime characterised in main-text §4.1. Panel (i) AR(2) shows the visibly tighter amplitude match consistent with its leading LR-EMD ranking under the matched-budget contract.

fail-systematically" framing of the artifact. The same JSON sidecar fields used to construct Figure A14 (`gen_log_delta_raw`, `drift_correction.gen_log_delta_mean_raw`, `drift_correction.shift_applied`) are sufficient to reproduce the un-shifted view for any of the nine models.

A.11 Per-Model Log-Return Distribution Diagnostics

Figures A16-A24 present, for each of the nine matched-budget generators (seed 42), a six-panel statistical-comparison grid on the log-return scale; the six-panel layout (PDF / CDF histograms, normal Q-Q, statistical moments, distance metrics, Shannon entropy comparison) is identical across all nine figures and is detailed in the caption of Figure A16. Sample construction matches the §A.10 protocol (777 contiguous log-deltas from each model's released sample bundle). The panels reflect the generator output distribution including any residual mean bias, and are visualisations of distributional fidelity at the log-return scale; they are not a restatement of the canonical matched-budget fidelity metrics of main-text §4.1 (which use a per-window evaluation protocol aggregated across all five seeds).

A.12 Per-Model DTW Alignment

Figure A25 provides per-model DTW alignment visualisations across all nine matched-budget generators: each subpanel shows the real log-delta sequence on the top axis (777 points), the generated log-delta sequence on the left axis (777 points, rotated), and the accumulated DTW warping-cost matrix as a heatmap in the centre with the optimal warping path drawn in red. The Sakoe-Chiba band is visible as the diagonal corridor; cells outside the band are excluded from the search.

Visualization protocol (differs from main-text §4.1). The cost matrix and warping path are computed via `dtaidistance.dtw.warping_paths(real, gen, window=500, psi=2)` followed by `dtaidistance.dtw.best_path`, then rendered via `dtaidistance.dtw_visualisation.plot_warpingpaths`. The displayed numerical DTW distance is therefore a visualization-protocol value: it

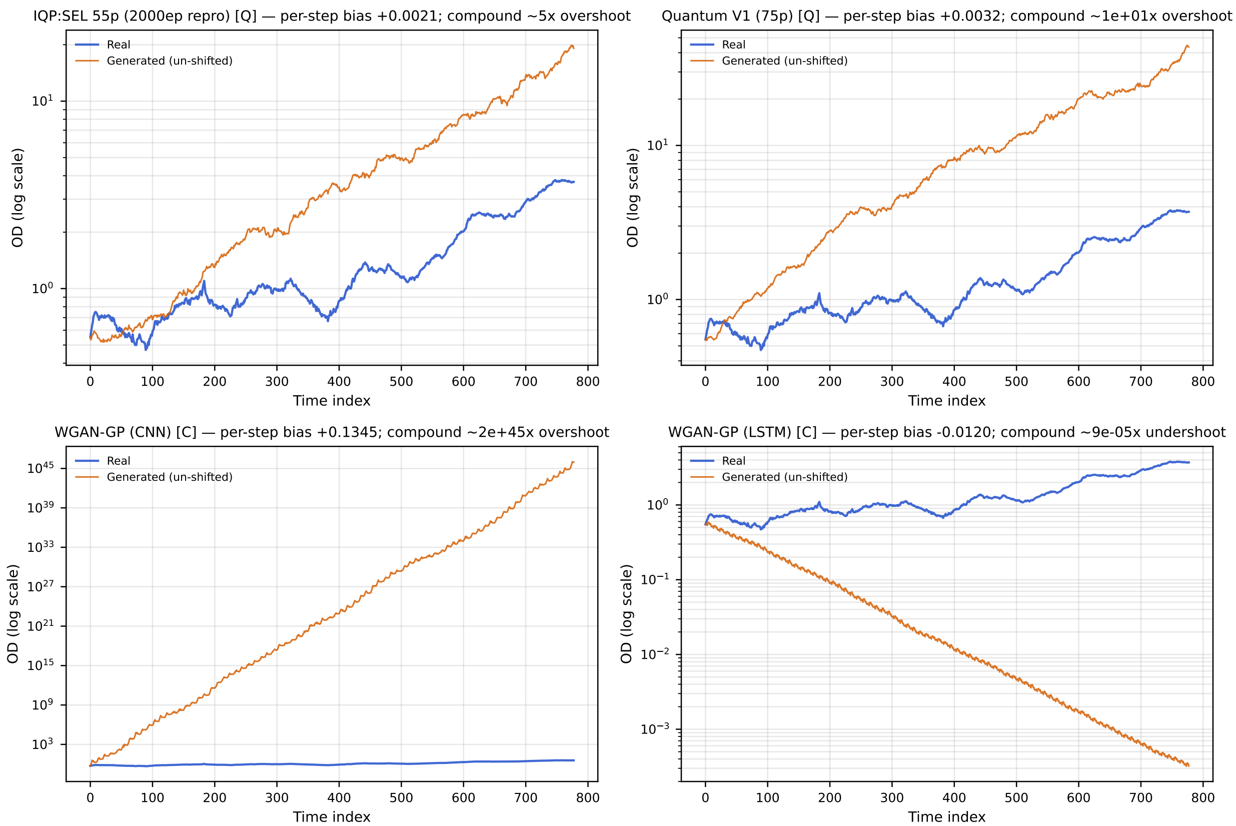


Figure A15: Un-shifted (raw) cumulative reconstructions for four representative models. Each panel: log-OD scale, real (blue) vs. un-corrected generated (orange) over 778 points. Per-panel title reports the per-step log-return bias and the compound factor at step 777 (`exp(777 · bias)`). Top row (quantum) shows $\sim$5-12$\times$ overshoot from per-step biases of $\sim$0.002-0.003 – the same artifact Figure A14 mean-shifts away to isolate structural fidelity. Bottom row shows the two classical models with the largest absolute per-step bias: WGAN-CNN (+0.135, compound $\sim 10^{45}$, off the top of the panel) and WGAN-LSTM (-0.012, compound $\sim 10^{-4}$, collapsing toward zero). The opposite-direction failure of these two classical models demonstrates that the compound-bias artifact is not architecture- specific and not directional; the mean-shift in Figure A14 removes it uniformly across all nine matched-budget models. The companion JSON for this figure records the per-step bias and compound factor per model in the `per_model` block.

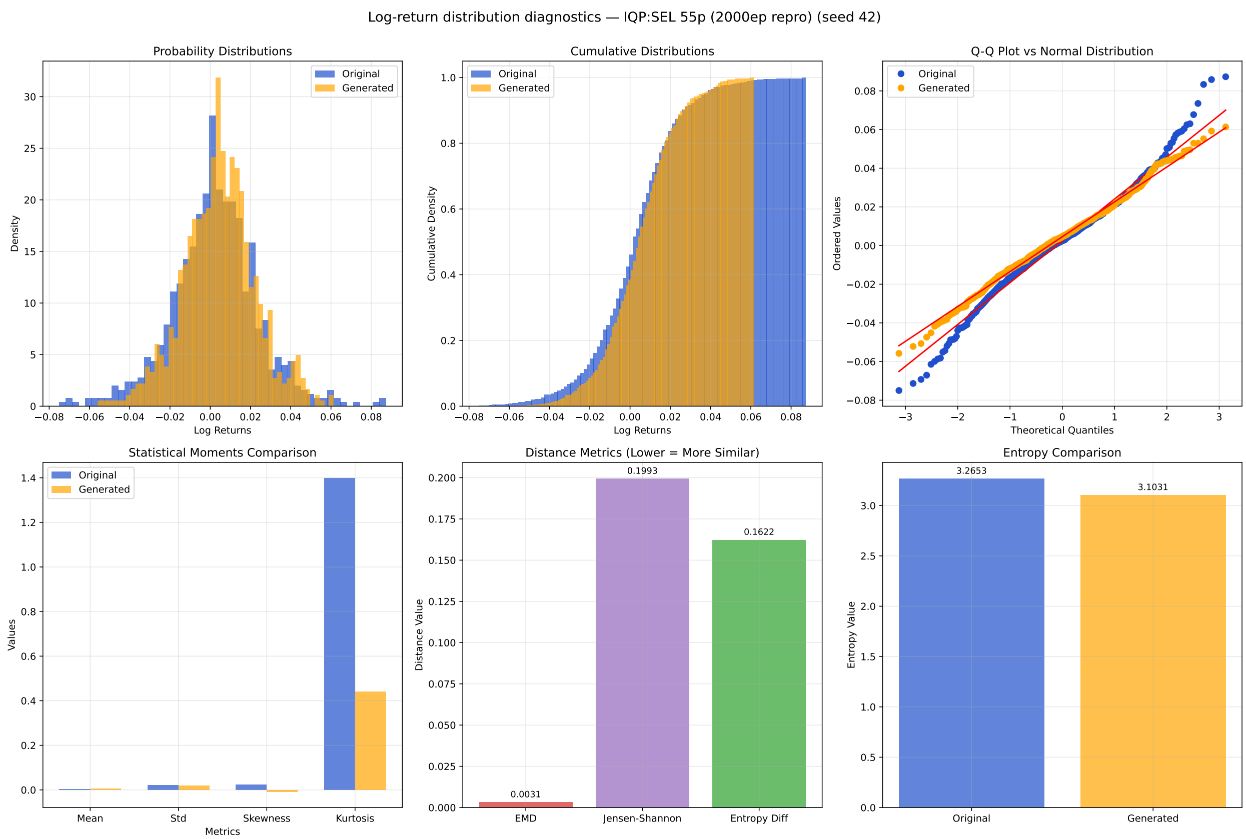


Figure A16: Log-return distribution diagnostics: IQP:SEL 55p (2000ep repro), seed 42. Real (blue) vs. generated (orange). Panel layout (identical across all nine §A.11 figures): top-left PDF histograms (50 bins, overlaid); top-middle CDF histograms (100 bins); top-right Q-Q vs. standard normal (deviation from the red reference line indicates non-Gaussianity); bottom-left moments bar chart (mean, std, skewness, Fisher excess kurtosis; std at ddof=0; kurtosis as `scipy.stats.kurtosis` default); bottom-middle distance metrics (EMD on raw samples; Jensen-Shannon on shared-bin histograms; $|\Delta H|$); bottom-right Shannon entropy comparison (50-bin histograms, $+10^{-12}$ per-bin floor).

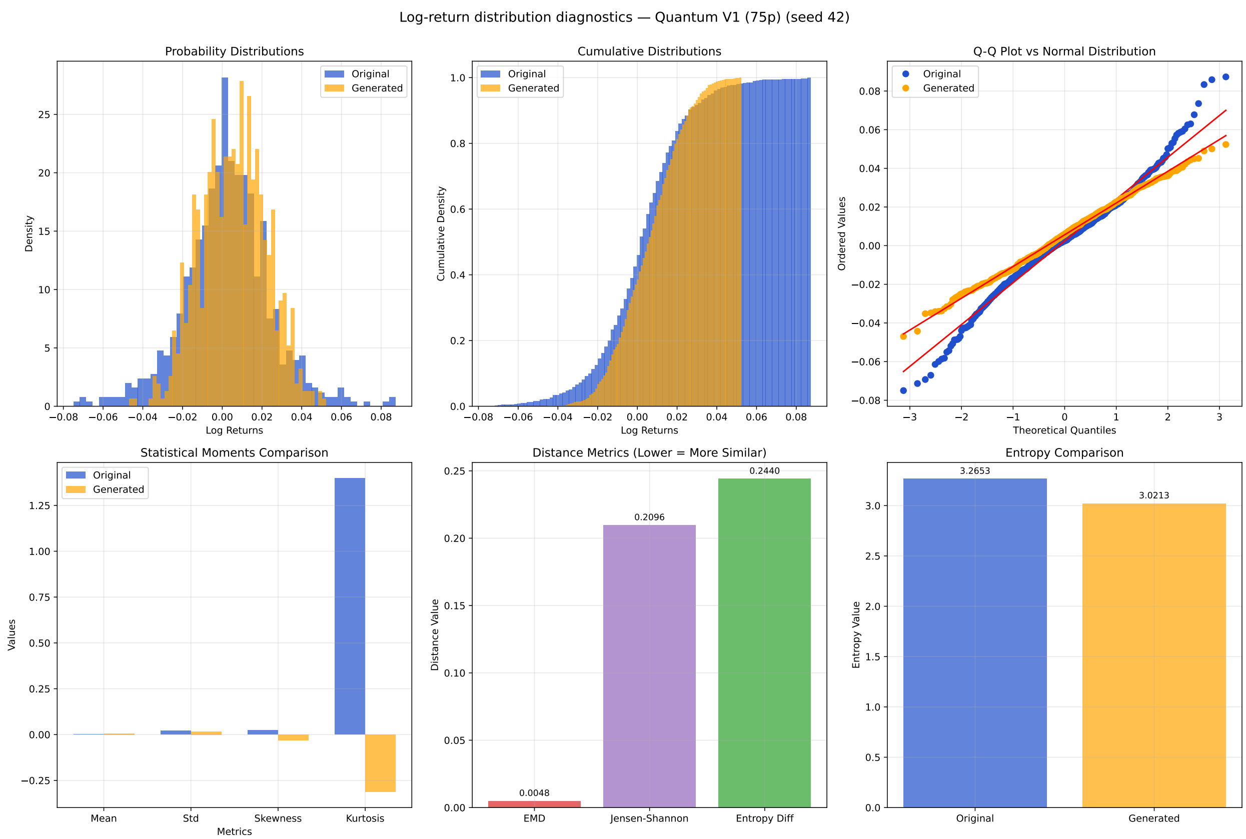


Figure A17: Log-return distribution diagnostics: Quantum V1 (75p), seed 42. Panels as in Figure A16.

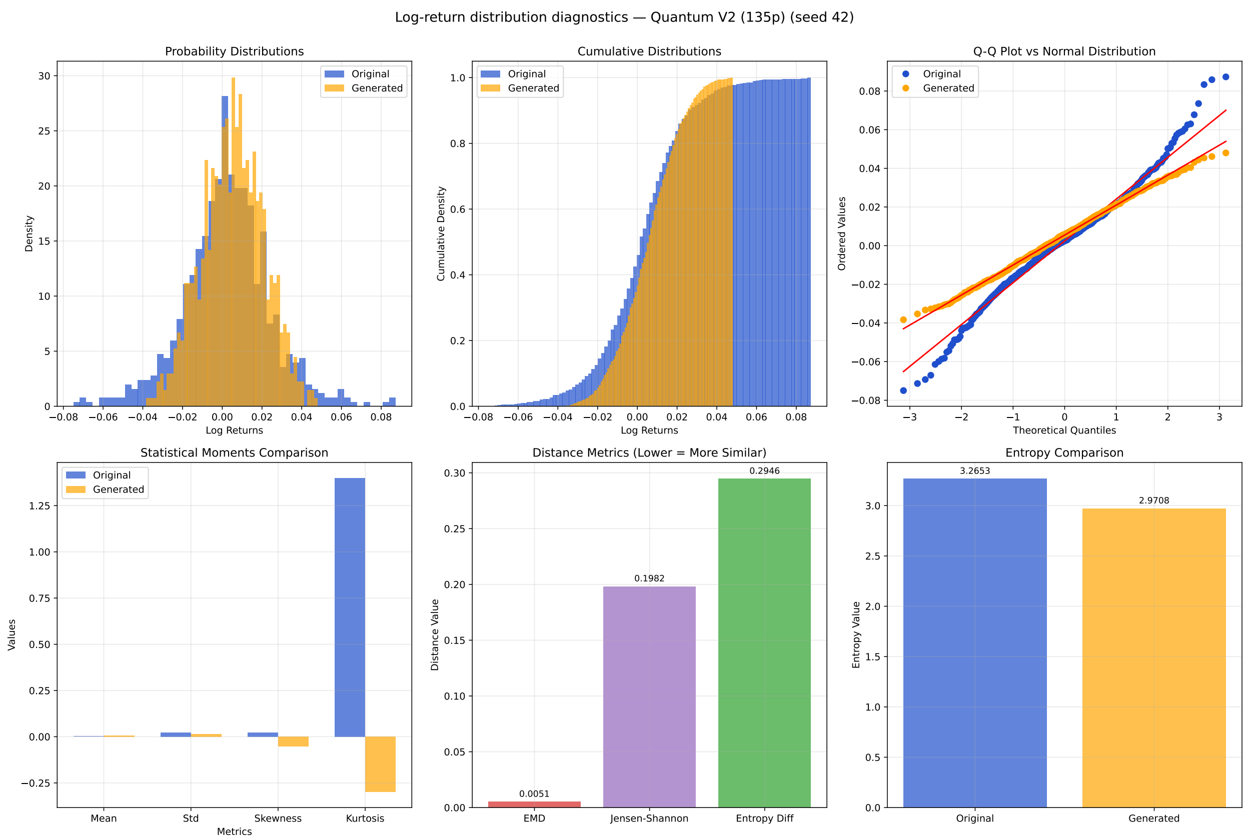


Figure A18: Log-return distribution diagnostics: Quantum V2 (135p), seed 42. Panels as in Figure A16.

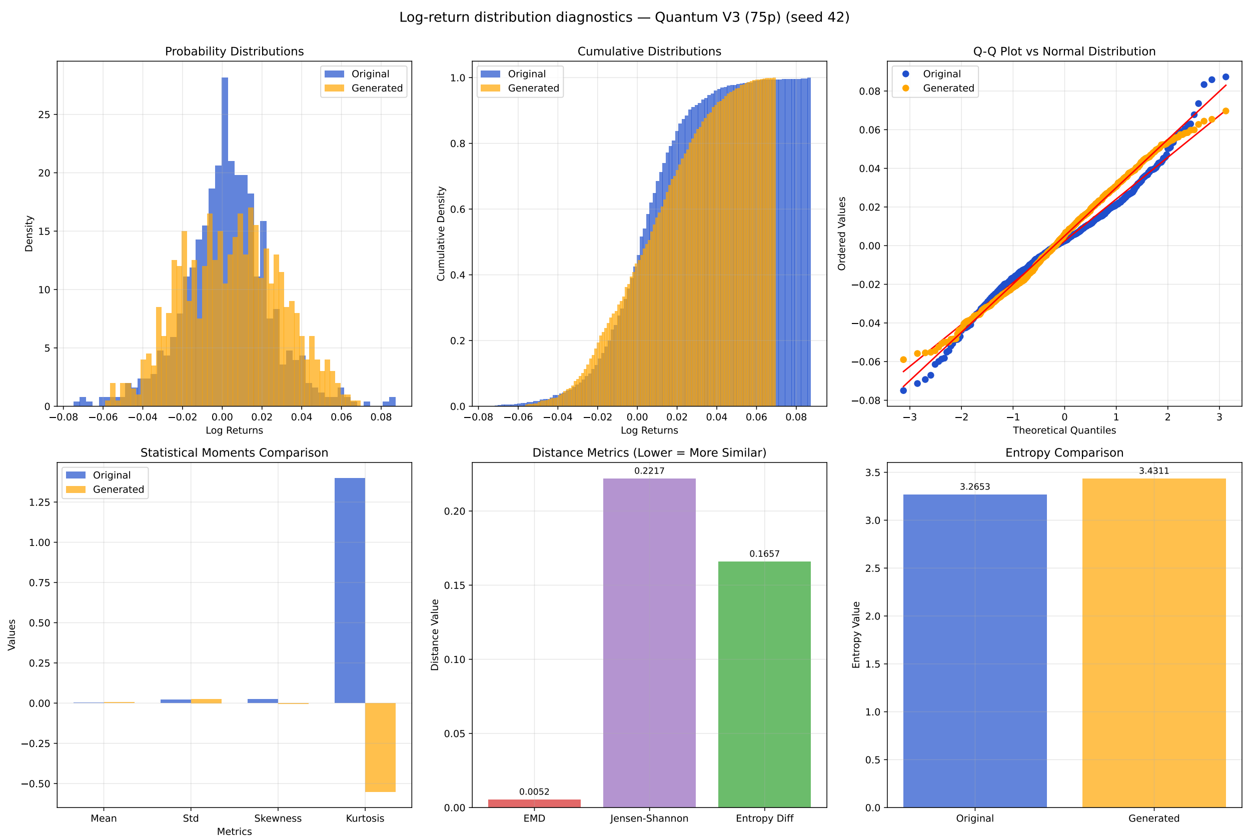


Figure A19: Log-return distribution diagnostics: Quantum V3 (75p, linear entangler), seed 42. Panels as in Figure A16.

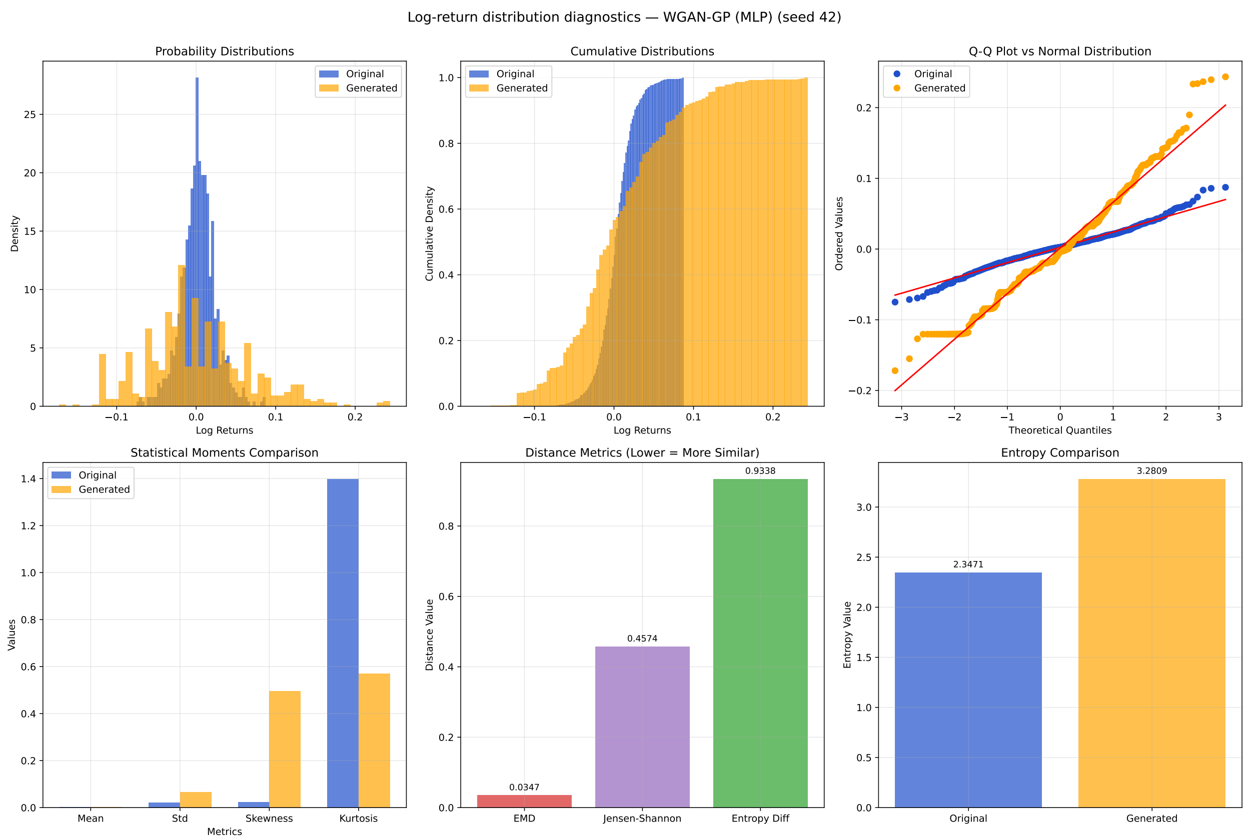


Figure A20: Log-return distribution diagnostics: WGAN-GP (MLP, 74p), seed 42. Panels as in Figure A16.

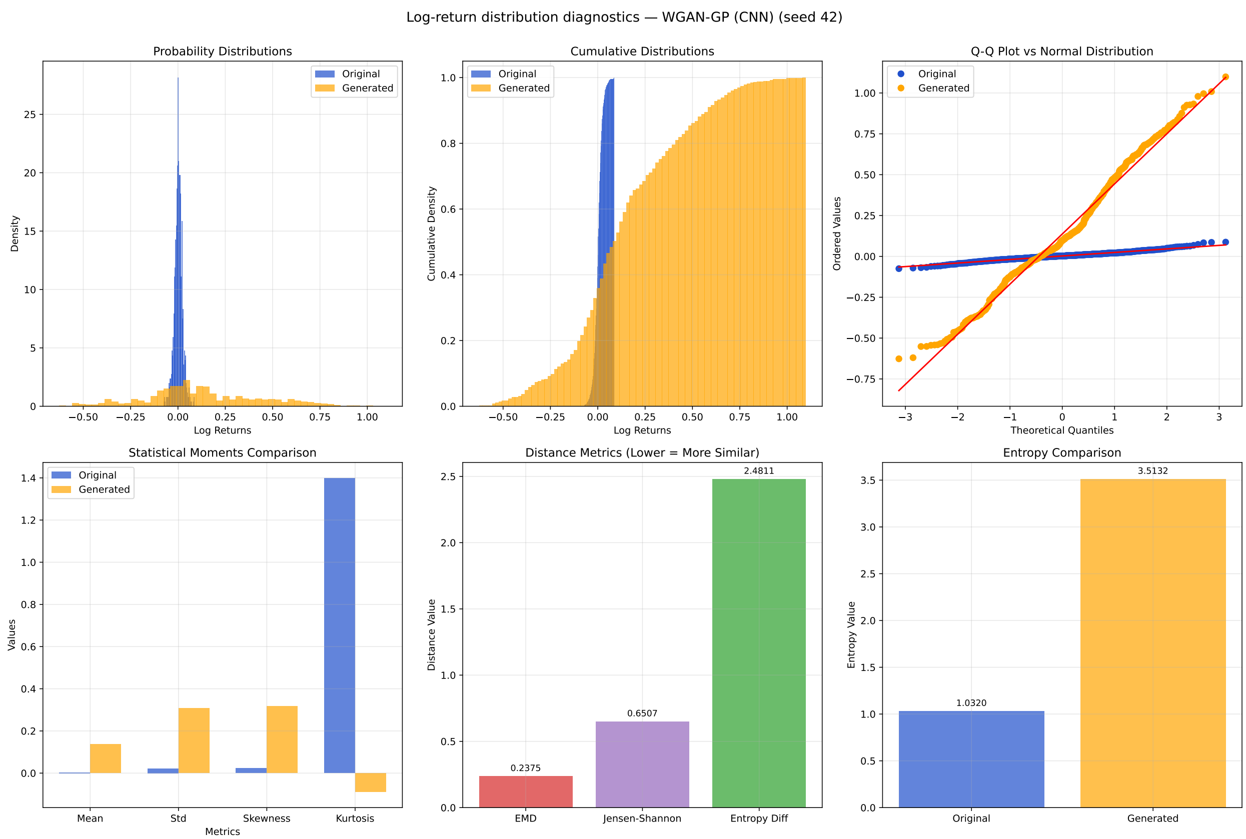


Figure A21: Log-return distribution diagnostics: WGAN-GP (CNN, 73p), seed 42 (the OD-EMD outlier seed disclosed in main-text §4.1). Panels as in Figure A16.

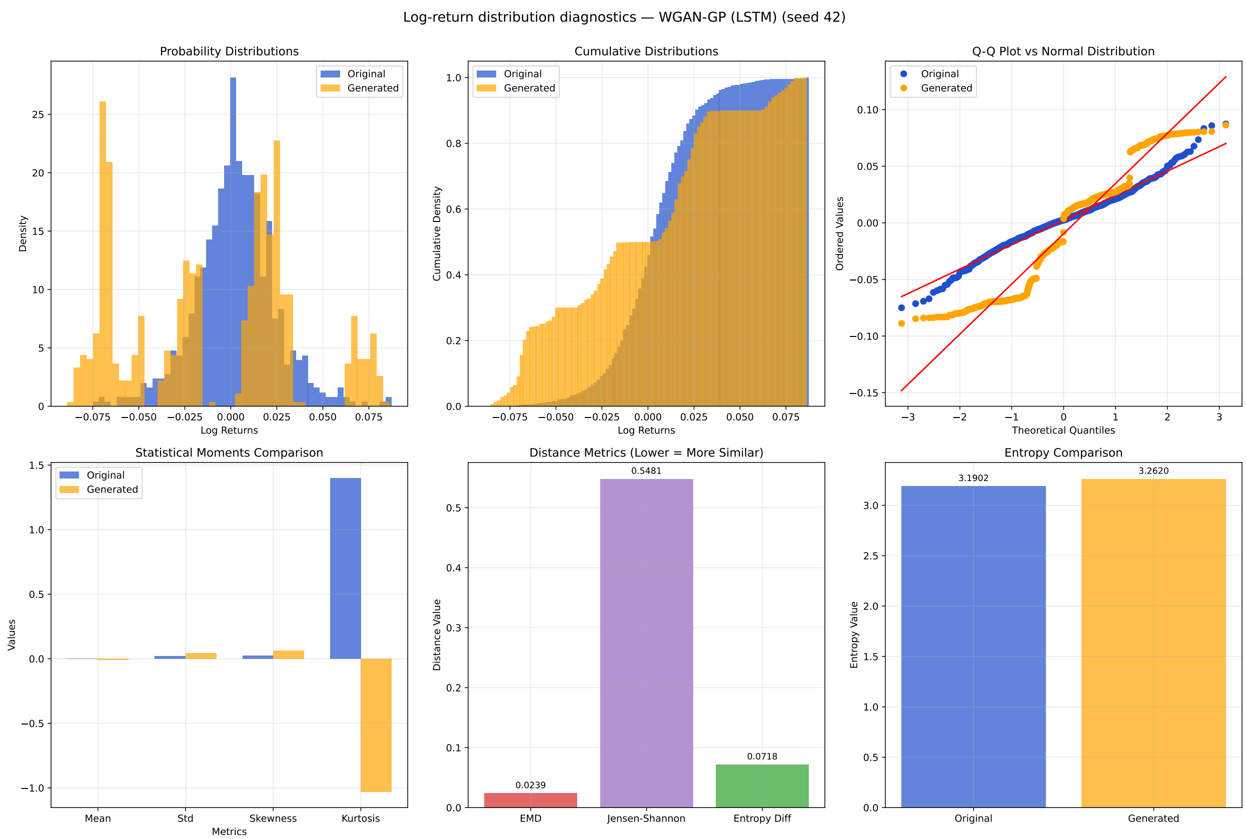


Figure A22: Log-return distribution diagnostics: WGAN-GP (LSTM, 78p), seed 42. Panels as in Figure A16.

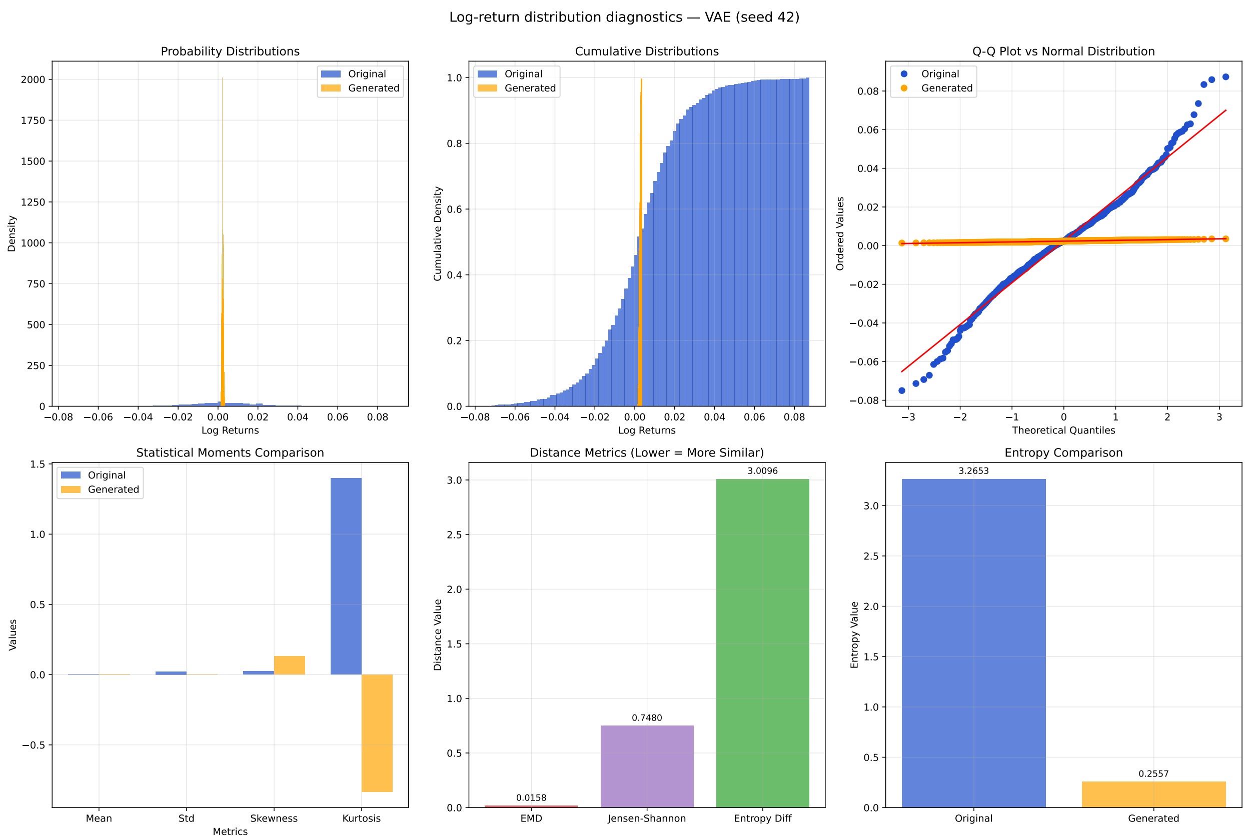


Figure A23: Log-return distribution diagnostics: VAE (562p, degenerate generation regime; see main-text §4.1), seed 42. Panels as in Figure A16.

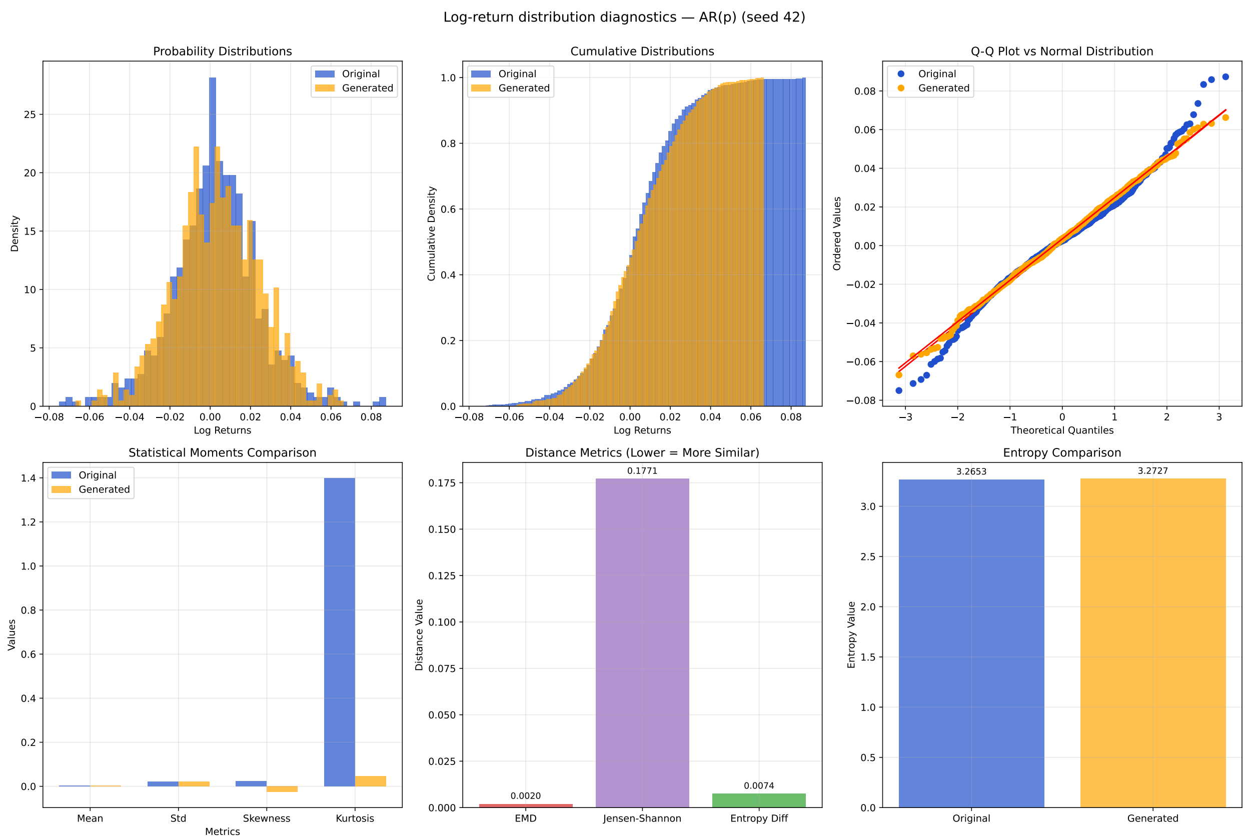
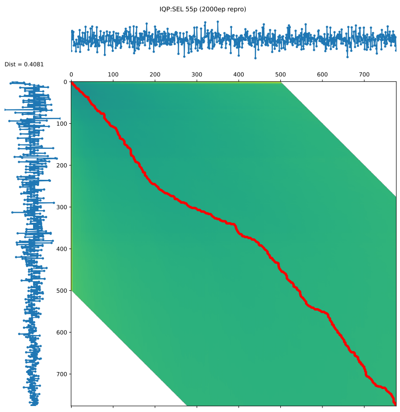


Figure A24: Log-return distribution diagnostics: AR(2) closed-form baseline, seed 42. Panels as in Figure A16. The visibly tightest amplitude/moment match in this collection is consistent with AR(2)'s leading LR-EMD ranking under the matched-budget contract reported in main-text §4.1.

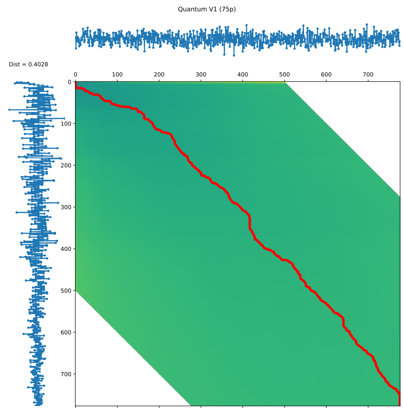
uses a different DTW implementation and band/skip-edge protocol than the canonical fastdtw-based LR-DTW reported in main-text §4.1 (which is computed under the matched-budget windowed-evaluation protocol per the `compute_dtw` entry point of C2). Numerical DTW distances in these figures are NOT directly comparable to the headline LR-DTW values; the figures are intended to illustrate alignment quality (warping-path shape, band utilization, anchor cost) rather than to re-derive the headline metric.

A.13 Quantum Hardware Robustness Sensitivity

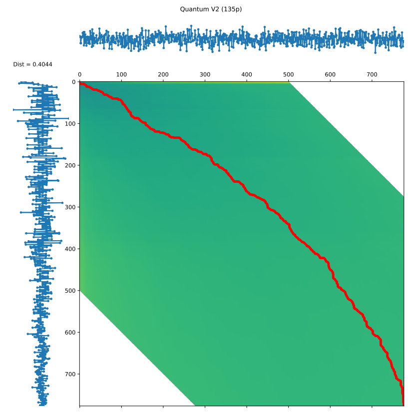
Reviewer R1-M5 requested an assessment of how the matched-budget finding would degrade under realistic quantum-hardware conditions (shot noise from finite-shot measurement; gate noise from depolarizing and amplitude-damping channels). Phase 12 of the revision provides two sensitivity studies addressed here: shot-noise sensitivity (1024 and 8192 shots vs. statevector analytic baseline) and noise-model sensitivity (depolarizing $p \in \{0, 0.001, 0.01, 0.05\}$ and amplitude-damping $\gamma \in \{0, 0.001, 0.01, 0.05\}$, per-layer channel insertion).



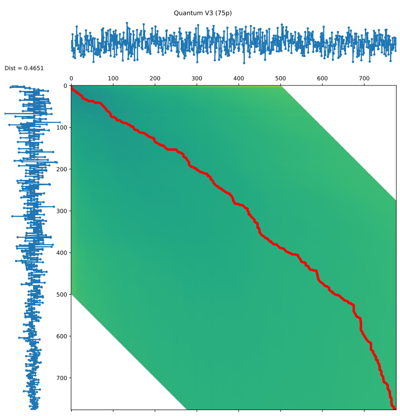
(a) IQP:SEL (55p, Q)



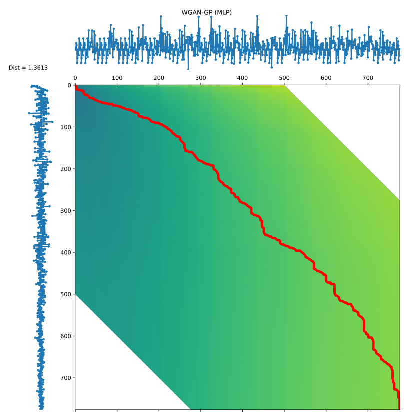
(b) V1 (75p, Q)



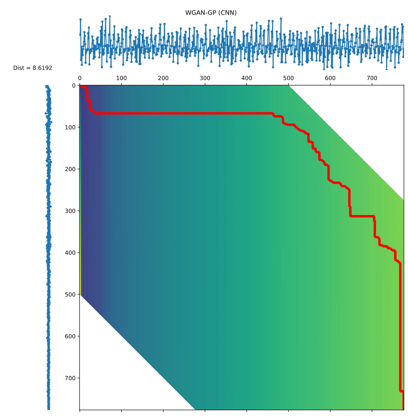
(c) V2 (135p, Q)



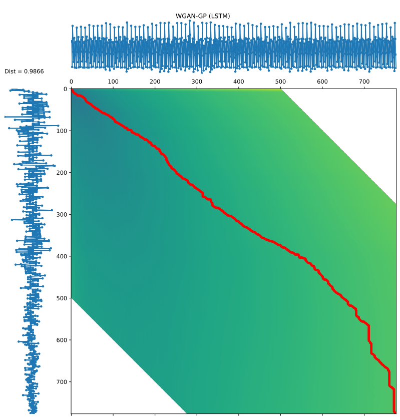
(d) V3 (75p, Q, linear entangler)



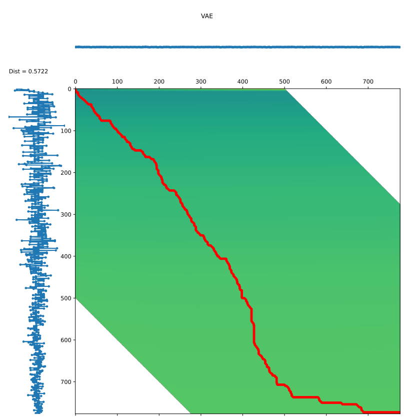
(e) WGAN-MLP (74p)



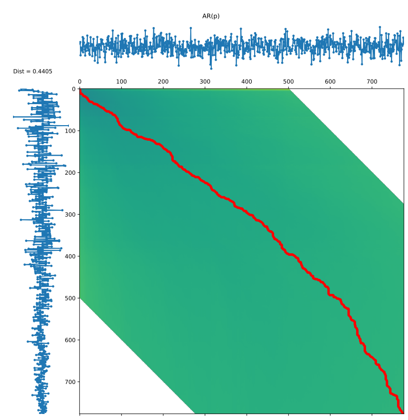
(f) WGAN-CNN (73p)



(g) WGAN-LSTM (78p)



(h) VAE (562p)



(i) AR(2) (3p)

Figure A25: Per-model DTW alignment atlas, seed 42, 2000ep matched budget. Each panel: top axis = real log-delta series; left axis = generated log-delta series; centre heatmap = accumulated DTW cost matrix (`dtaidistance`, `window=500`, `psi=2`); optimal warping path in red. Distance values displayed in each panel are visualisation-protocol values (see section intro) and are NOT directly comparable to the headline fastdtw-based LR-DTW reported in main-text §4.1. Panel (f) is the WGAN-CNN seed-42 OD-EMD outlier disclosed in main-text §4.1; panel (h) is the VAE degenerate-generation regime characterised in main-text §4.1.

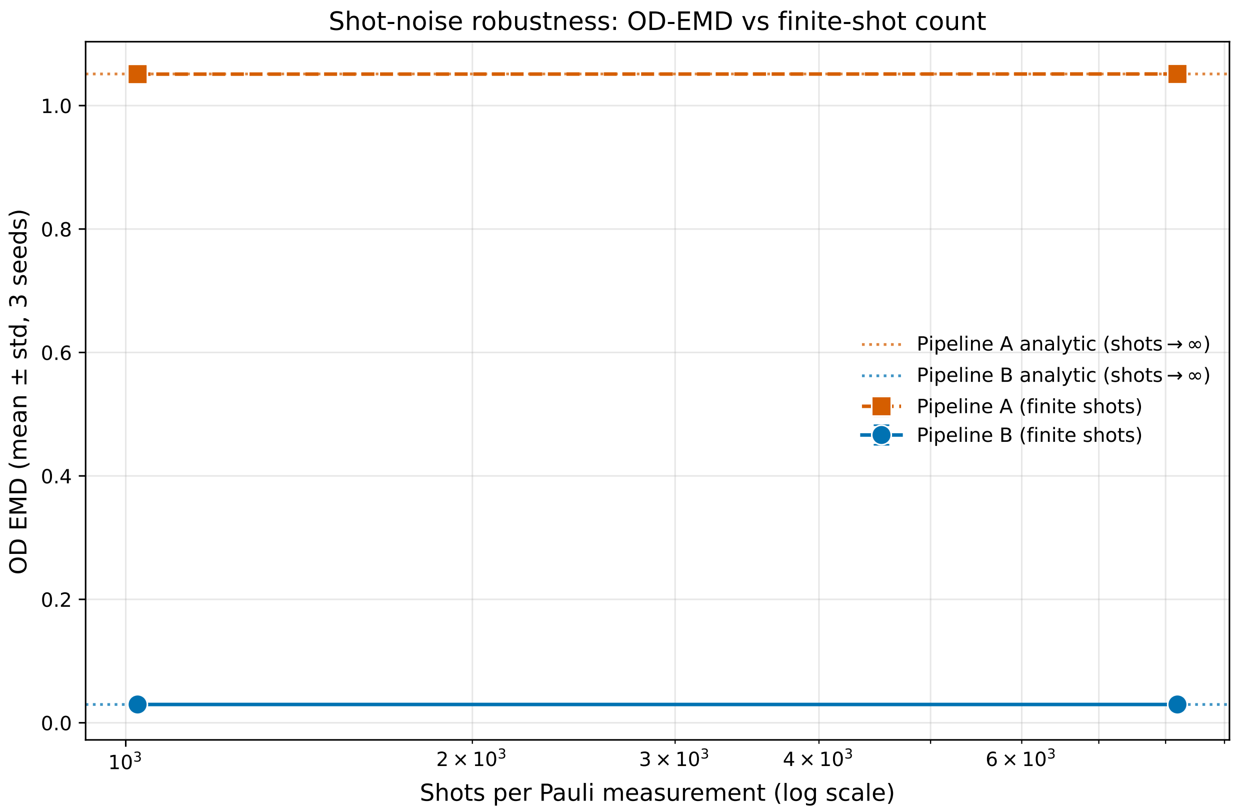


Figure A26: Shot-noise sensitivity of the quantum generator (IQP:SEL 55p) under finite-shot measurement: analytic statevector baseline vs. 8192 shots vs. 1024 shots. OD-EMD reported as 3-seed mean $\pm$ sample-std (seeds $\{42, 43, 44\}$) on Pipelines A and B. The 1024-shot and 8192-shot curves remain within sampling noise of the analytic baseline, indicating the matched-budget finding is robust to the shot-noise levels accessible on near-term hardware.

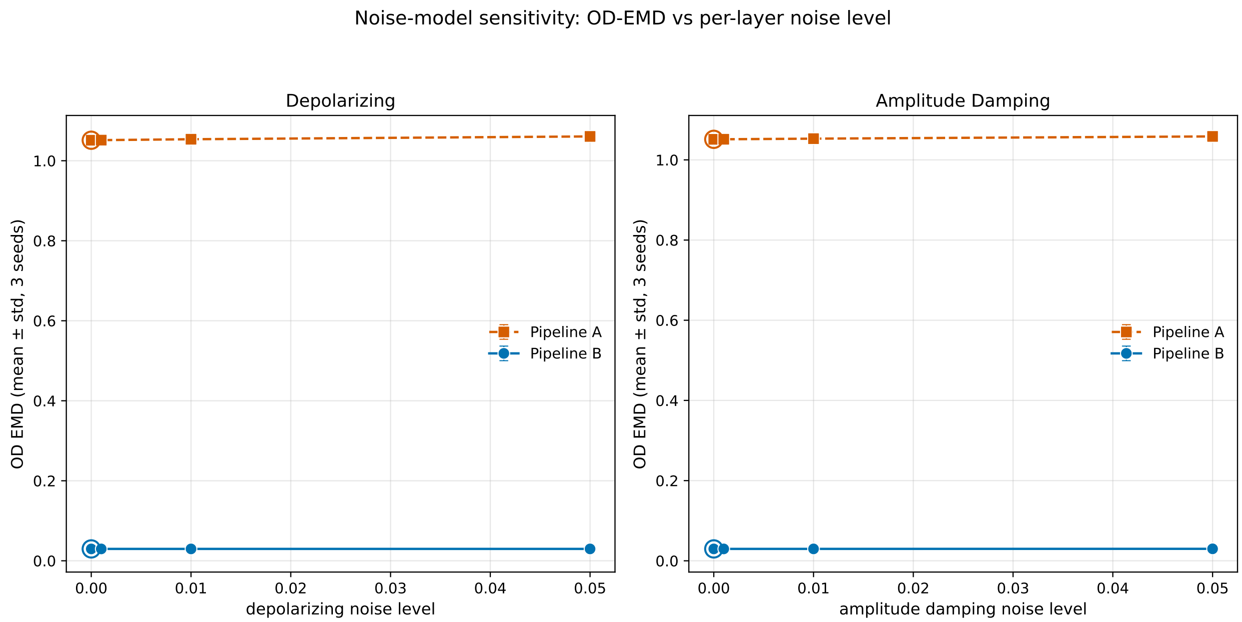


Figure A27: Noise-model sensitivity of the quantum generator (IQP:SEL 55p) under per-layer depolarizing and amplitude-damping channels (inserted after each entangling block). OD-EMD as 3-seed mean $\pm$ sample-std (seeds $\{42, 43, 44\}$) at noise levels $\{0, 0.001, 0.01, 0.05\}$ per channel type. Pipelines A and B shown as separate curves. Both channels show graceful degradation: the matched-budget finding is preserved at the 0.001 level, begins to degrade at 0.01, and is degraded at 0.05; consistent with the noise tolerance reported for similar PQC variational circuits in the literature.